Controller Design for Flapping Flight

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1 Introduction

We propose a control strategy for flapping flight.

2 Model Description

2.1 Overview

The flapping flight dynamics model consists of a point mass subjected to its own weight $m\vec{g}$ and to the aerodynamic force $\vec{F}$ from N wing pairs. The point mass is free to move in the (x, z) plane, with the spatial direction $\hat{z}$ such that $\vec{g} = -g\hat{z}$, and $\hat{x}$ such that $(\hat{x}, \hat{z})$ forms a direct orthonormal basis. The body position is $\vec{x} = x\hat{x} + z\hat{z}$ and the body velocity is $\vec{u} = \dot{x}\hat{x} + \dot{z}\hat{z}$.

2.2 Aerodynamic Force

The total wing area is A, and both wings of each pair are lumped into a single, massless surface element with area A/N. We will refer to these surface elements as *wing elements*, keeping in mind they each represent *both* wings of a left-right pair. The aerodynamic properties of a wing element are represented by the quasi-steady model of [1, 2]:

$$C_L(\alpha) = C_{L,0} \sin 2\alpha \tag{1}$$

$$C_D(\alpha) = C_{D,0} \cos^2 \alpha + C_{D,\frac{\pi}{2}} \sin^2 \alpha \tag{2}$$

where α is the angle of attack, and the parameter values are $C_{L,0} = 1.5$, $C_{D,0} = 0.1$, and $C_{D,\frac{\pi}{2}} = 2.0$. The model neglects any aerodynamic interaction between wing elements. To express the wing element kinematics and aerodynamic force $\vec{F}$, it is convenient to introduce the stroke basis $(\hat{s}, \hat{n})$, obtained by rotating the fixed frame basis $(\hat{x}, \hat{z})$ by the *stroke plane angle* γ. That is

$$\hat{s} = \cos\gamma\hat{x} + \sin\gamma\hat{z} \tag{3}$$

$$\hat{n} = -\sin\gamma\hat{x} + \cos\gamma\hat{z} \tag{4}$$

Each wing element translates along the stroke axis $\hat{s}$, and $\hat{n}$ represents the stroke plane normal. The wing element position relative to the point mass body is $\vec{s} = s(t)\hat{s}$, where $s(t)$ is the prescribed flapping coordinate. Because the wing is instantaneously at a distance $s(t)$ from the body, the aerodynamic force acts on the body via a moment arm $s(t)$. However, in the present point mass model, there is no rotational degree of freedom, and the force is effectively considered to act through the center of mass.

Supposing γ stays constant during the wing stroke, the wing velocity relative to the body is $\dot{\vec{s}} = \dot{s}(t)\hat{s}$. With the body velocity in the fixed frame (relative to still air) $\vec{u}$, the wing velocity relative to the air is $\vec{v} = \dot{\vec{s}} + \vec{u}$. The aerodynamic force for wing pair i is given below (note the subscript i is omitted from quantities on the right hand side for tidiness)

$$\vec{F}_i = \frac{1}{2}\rho\frac{A}{N}|\vec{v}|^2 \left[C_L(\alpha)\hat{\ell} + C_D(\alpha)\hat{d}\right] \tag{5}$$

where ρ is the air density and the drag direction vector $\hat{d}$ is defined as

$$\hat{d} = -\hat{v} = -\frac{\vec{v}}{|\vec{v}|} \tag{6}$$

and the lift direction vector $\hat{\ell}$ is defined as

$$\hat{\ell} = -\hat{d}^{\perp} \tag{7}$$

where $\cdot^{\perp}$ denotes counterclockwise rotation by $\pi/2$. As a result, $(\hat{\ell}, \hat{d})$ forms a direct orthonormal basis. To express the angle of attack α, it is convenient to define the chord direction vector $\hat{c}$, pointing from the trailing edge to the leading edge. Let $\psi(t)$ be the wing pitch angle, relative to the stroke plane normal $\hat{n}$. The chord direction vector is

$$\hat{c} = -\sin\psi\,\hat{s} + \cos\psi\,\hat{n} \tag{8}$$

The geometry introduced so far is illustrated in Figure 1. The angle of attack α can be defined here as the

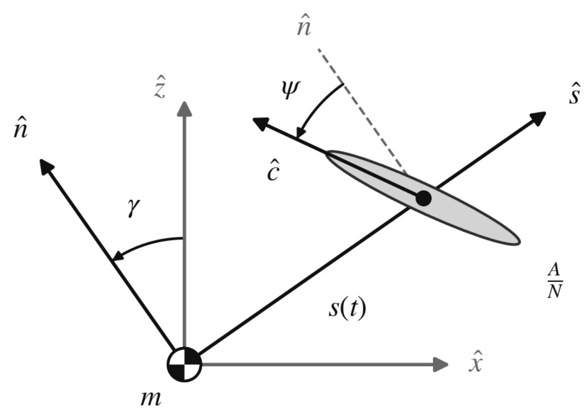

Figure 1: Model geometry. The stroke basis $(\hat{s}, \hat{n})$ is obtained by rotating the fixed frame basis $(\hat{x}, \hat{z})$ by the stroke plane angle γ. The wing element sits at position $s(t)\,\hat{s}$ relative to the point mass body, with chord direction $\hat{c}$ at pitch angle ψ from the stroke plane normal $\hat{n}$. For simplicity, only one wing element is drawn.

signed angle from the velocity direction vector $\hat{v}$ to the chord direction vector $\hat{c}$. We show below that this definition is consistent with the definition of the lift direction vector $\hat{\ell}$, giving the correct sign for lift. With this definition,

$$\cos\alpha = \hat{v} \cdot \hat{c} \tag{9}$$

$$\sin\alpha = \hat{v} \times \hat{c} \tag{10}$$

where $\hat{v} = \vec{v}/|\vec{v}|$ and $\times$ denotes the 2D cross product, defined as $\vec{a} \times \vec{b} = a_1 b_2 - a_2 b_1$. $C_L(\alpha)$ can be rewritten as a function of $\cos\alpha$ and $\sin\alpha$ only, indeed

$$\begin{aligned} C_L(\alpha) &= C_{L,0} \sin 2\alpha \\ &= 2C_{L,0} \sin\alpha \cos\alpha \end{aligned} \tag{11}$$

C_L and C_D can then be expressed with simple vector math as follows

$$C_L = 2C_{L,0}(\hat{v} \cdot \hat{c})(\hat{v} \times \hat{c}) \tag{12}$$

$$C_D = C_{D,0}(\hat{v} \cdot \hat{c})^2 + C_{D,\frac{\pi}{2}}(\hat{v} \times \hat{c})^2 \tag{13}$$

The lift and drag directions resulting from the above expressions for C_L, C_D, $\hat{\ell}$, and $\hat{d}$ are checked for correctness in all four quadrants of the angle of attack in Figure 2, generated directly from the expressions. Because $\hat{\ell}$ and $\hat{d}$ are constructed from $\hat{v}$ alone, the wing attitude enters only through the scalar coefficients, so it suffices to check the coefficient signs quadrant by quadrant. Naturally, drag is positive along $\hat{d}$ for every α. The sign of C_L is that of $(\hat{v} \cdot \hat{c})(\hat{v} \times \hat{c}) = \cos\alpha \sin\alpha$: positive in quadrants I and III, negative in II and IV, as drawn.

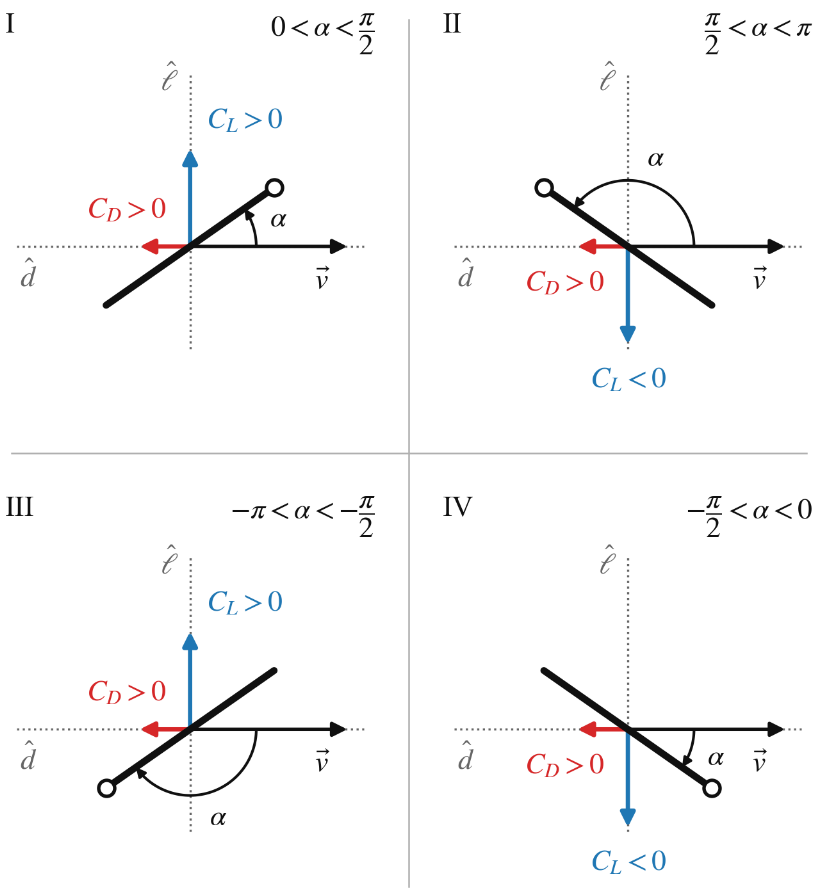

Figure 2: Lift and drag directions in the four quadrants of the angle of attack α, measured from the wing velocity direction $\hat{v}$ to the chord direction $\hat{c}$. Panels are labeled by quadrant (I–IV). Drag ($C_D > 0$) always opposes the wing velocity along $\hat{d}$, while the sign of C_L orients lift along $\pm\hat{\ell}$. The hollow circle marks the wing leading edge (the $\hat{c}$ end of the chord).

2.3 Wing Kinematics

Wing motion is prescribed by the flapping coordinate $s(t)$ and the pitch angle $\psi(t)$. Flapping motion for each wing element is taken to be sinusoidal:

$$s(t) = s_0 \cos(\omega t + \phi) \tag{14}$$

where s_0 and ω are set parameters for all wing pairs, and ϕ is a phase parameter specific to a given wing pair, allowing for out of phase flapping of two wing pairs. For pitch, we consider two different kinematics. The first is sinusoidal:

$$\psi(t) = \psi_0 + \psi_1 \cos(\omega t + \phi - \delta_0) \tag{15}$$

where ψ_0 (mean pitch), ψ_1 (pitch amplitude), and δ_0 (pitch phase lead) are set parameters for all wing pairs. The second kinematics law is simpler: through each half of the wing stroke, the wing pitch remains constant. The first half of the stroke, when the wing moves along $-\hat{s}$, corresponds to $\omega t + \phi \in [0, \pi[\pmod{2\pi}$. The second half, when the wing moves along $+\hat{s}$, corresponds to $\omega t + \phi \in [\pi, 2\pi[\pmod{2\pi}$. The simplified pitch kinematics are, then,

$$\psi(t) = \begin{cases} \psi_0 + \psi_1 & \text{if } \omega t + \phi \in [0, \pi[\pmod{2\pi} \\ \psi_0 - \psi_1 & \text{if } \omega t + \phi \in [\pi, 2\pi[\pmod{2\pi} \end{cases} \tag{16}$$

This can be considered an approximation of the sinusoidal law with $\delta_0 = \frac{\pi}{2}$. Both pitch kinematics laws are illustrated in Figure 3. As the figure makes apparent, $\delta_0 = \frac{\pi}{2}$ keeps the leading edge oriented into the wing motion throughout the cycle, with pitch extremes at mid-stroke and $\psi = \psi_0$ at stroke reversal. $\delta_0 = -\frac{\pi}{2}$ produces the opposite, adverse orientation, while $\delta_0 = 0$ and π place the pitch extremes at the reversals.

2.4 Equation of Motion and Nondimensionalization

The point mass is subjected to its own weight under gravity, and to the aerodynamic force from each wing pair, so the equation of motion is

$$m\frac{d^2\vec{x}}{dt^2} = m\vec{g} + \frac{1}{2}\rho A \sum_{i=1}^{N} \frac{1}{N}|\vec{v}_i|^2 \left[C_{L,i}\hat{\ell}_i + C_{D,i}\hat{d}_i\right] \tag{17}$$

The equation can be nondimensionalized using the body length L as the length scale, and the gravitational time scale $\sqrt{L/g}$, giving an 'acceleration scale' g and a 'velocity scale' $\sqrt{gL}$. Introducing the $*$ superscript to denote nondimensionalized quantities, the equation of motion can be divided by mg and becomes

$$\begin{aligned} \ddot{x}^*\hat{x} + \ddot{z}^*\hat{z} &= -\hat{z} + \frac{1}{2}\frac{\rho A L}{m}\frac{1}{N}\sum_{i=1}^{N}\frac{1}{gL}|\vec{v}_i|^2 \left[C_{L,i}\hat{\ell}_i + C_{D,i}\hat{d}_i\right] \\ &= -\hat{z} + \frac{1}{2}\frac{\rho A L}{m}\frac{1}{N}\sum_{i=1}^{N}|\vec{v}_i^*|^2 \left[C_{L,i}\hat{\ell}_i + C_{D,i}\hat{d}_i\right] \end{aligned} \tag{18}$$

We introduce the dimensionless inverse wing loading A^*/m^* defined as

$$\frac{A^*}{m^*} = \frac{\rho A L}{m} \tag{19}$$

giving the final form of the dimensionless equation of motion

$$\ddot{x}^*\hat{x} + (\ddot{z}^* + 1)\hat{z} = \frac{1}{2}\frac{A^*}{m^*}\frac{1}{N}\sum_{i=1}^{N}|\vec{v}_i^*|^2 \left[C_{L,i}\hat{\ell}_i + C_{D,i}\hat{d}_i\right] \tag{20}$$

The wing kinematics themselves must also be nondimensionalized. Introducing $\tau^* = \omega t$ and $s_0^* = s_0/L$,

$$s^*(\tau^*) = s_0^* \cos(\tau^* + \phi) \tag{21}$$

$$\psi(\tau^*) = \psi_0 + \psi_1 \cos(\tau^* + \phi - \delta_0) \tag{22}$$

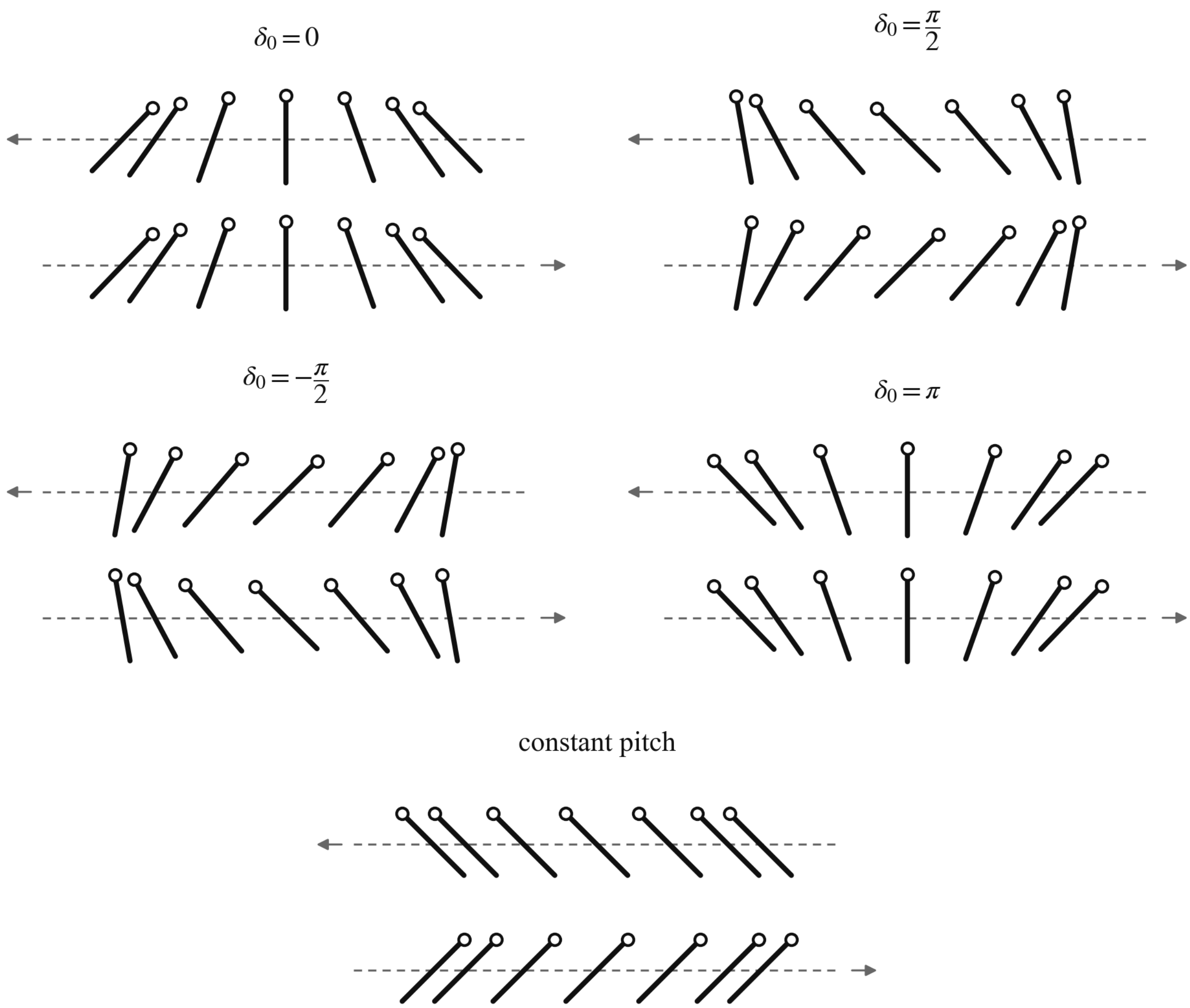

Figure 3: Wing motion through the two halves of the flapping cycle, for the sinusoidal pitch kinematics with $\delta_0 \in \{0, \frac{\pi}{2}, -\frac{\pi}{2}, \pi\}$ and for the constant pitch simplification. Across all panels, $\psi_0 = 0$ and $\psi_1 = 45°$. The top row shows the first half-stroke (wing moving along $-\hat{s}$, with $\hat{s}$ pointing right) and the bottom row the second half-stroke, along $+\hat{s}$. Arrows indicate the direction of motion, the hollow circle marks the wing leading edge, and snapshots are taken at equal time increments, crowding near the stroke reversals where the flapping motion slows. The constant pitch kinematics match the $\delta_0 = \pi/2$ kinematics well, except near stroke reversal.

Finally, the dimensionless wing velocity along $\hat{s}$ relative to the body is

$$\begin{aligned}
\dot{s}^*(\tau^*) &= \frac{1}{\sqrt{gL}} \frac{d}{dt} s_0 \cos(\omega t + \phi) \\
&= -\frac{1}{\sqrt{gL}} s_0 \omega \sin(\tau^* + \phi) \\
&= -s_0^* \omega^* \sin(\tau^* + \phi)
\end{aligned} \tag{23}$$

with $\omega^* = \omega\sqrt{L/g}$.

3 Controller Design

3.1 Problem Statement

We wish to design a control strategy to achieve arbitrary trajectories in the two-dimensional plane, including prey pursuit trajectories. We assume that the flyer's own state is known using a variety of sensory inputs, so position and velocity are available as inputs to the controller. In addition, when a prey is present in the environment, the bearing angle to the prey, as detected by the eyes, is also an available input. Because we are particularly interested in *dragonfly* flight, we make two design decisions such that the resulting flight characteristics resemble those of a dragonfly. First, hovering should be achieved with an inclined stroke plane (tilted from the horizontal), characterized by the parameter γ^{hover}. Second, when at speed, the stroke-plane normal should be approximately aligned with the direction of travel, as observed in experiments [3]. This last choice will simplify the analysis to follow, since any flow velocity component parallel to the stroke plane can be neglected. With these choices in mind, the contents of this section otherwise hold for flapping flight in general, as we do not assume specific morphological characteristics like number of wing pairs, size, etc.

3.2 Mean Aerodynamic Force

3.2.1 Definitions and Assumptions

To affect its trajectory, the flying body must modulate and direct the aerodynamic force acting on its wings. Because the aerodynamic force on a flapping wing is by nature highly unsteady, it is convenient to consider the wingbeat-cycle average of the aerodynamic force, summed over each of the wings, which we call the *mean aerodynamic force*, $\langle \vec{F}^* \rangle$. $\langle \vec{F}^* \rangle$ is a function of morphology and wing kinematics, as well as the flow characteristics in the body frame (this involves the body kinematics in the fixed frame, as well as any wind in the fixed frame; in the following we refer to this simply as the body kinematics, or body velocity, with the implication that it is relative to the air, which need not be the same as the fixed frame if there is any wind). Body velocity is not constant over a wingbeat cycle: there are (a) oscillations inherent to flapping, and there may be (b) a net change in body velocity over the cycle. Assuming the oscillations are small compared to the characteristic wing velocity, and that any body velocity changes occur at a slow rate relative to the wingbeat period, the body velocity may be considered constant over a wingbeat cycle. The following assumes these conditions are met. Our objective is now to understand how $\langle \vec{F}^* \rangle$ depends on the wing kinematics at any given body velocity, so that we are able to formulate a control strategy.

As explained in subsection 3.1, we are designing a controller under the assumption that the body velocity is normal to the stroke plane. Beyond morphology and kinematics, then, there is a single body velocity parameter $\langle u^* \rangle$, which is the cycle-averaged body velocity magnitude (the direction being set to the stroke-plane normal $\hat{n}$). It is convenient to introduce a similarity parameter in the form of the *advance ratio*, J, which is the ratio of the cycle-averaged body velocity to the characteristic wing velocity:

$$J = \frac{\langle u^* \rangle}{s_0^* \omega^*} \tag{24}$$

The instantaneous wing velocity $\vec{v}^*$ relative to the air has the stroke-plane normal and in-plane components

$$v_n^* = \langle u^* \rangle = s_0^* \omega^* J \tag{25}$$
$$v_s^* = -s_0^* \omega^* \sin \tau^* \tag{26}$$

with τ^* as introduced in section 2; the wing-pair phase ϕ is dropped without loss of generality, since the cycle averages below are unaffected by it. The $+\hat{s}$ half-stroke corresponds to $\sin \tau^* < 0$. The squared norm of the velocity is

$$(v^*)^2 = \left(s_0^* \omega^*\right)^2 \left[J^2 + \sin^2 \tau^*\right] \tag{27}$$

We introduce the signed angle χ from $\hat{n}$ to $\vec{v}^*$, defined as

$$\chi = -\tan^{-1} \frac{v_s^*}{v_n^*} = \tan^{-1} \frac{\sin \tau^*}{J} \tag{28}$$

Both ψ and χ are signed angles measured from $\hat{n}$ with the same sense (positive tilting toward $-\hat{s}$), locating the chord direction $\hat{c}$ and the velocity direction $\hat{v}$ respectively. The angle of attack of section 2 — the signed angle from $\hat{v}$ to $\hat{c}$ — is therefore simply

$$\alpha = \psi - \chi = \psi - \tan^{-1} \frac{\sin \tau^*}{J} \tag{29}$$

valid on both half-strokes, with no case distinction on the stroke direction. The above is sufficient to obtain the mean aerodynamic force. The derivation is somewhat involved in the general case. We first present the hovering case, which corresponds to $J = 0$.

3.2.2 Hover Case ($J = 0$)

For $J = 0$, the flow angle saturates at $\chi = \frac{\pi}{2} \frac{\sin \tau^*}{|\sin \tau^*|}$ and the angle of attack simplifies to

$$\alpha = \psi - \frac{\pi}{2} \frac{\sin \tau^*}{|\sin \tau^*|} \tag{30}$$

The four sign combinations of $(\sin \tau^*, \psi)$ place α one in each quadrant of Figure 2. Using the appropriate trigonometric identities, the aerodynamic coefficients (defined in Equation 1 and Equation 2) are

$$C_L = -C_{L,0} \sin 2\psi \tag{31}$$
$$C_D = \frac{C_{D,\frac{\pi}{2}} + C_{D,0}}{2} + \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \cos 2\psi \tag{32}$$

both independent of the stroke direction; the direction dependence resides entirely in the force direction vectors. Because the wing velocity relative to the air is purely in-plane (along $\hat{s}$), the in-plane aerodynamic force component is purely drag, and the out-of-plane component (along $\hat{n}$) is purely lift: the force directions are $\hat{d} = \frac{\sin\tau^*}{|\sin\tau^*|}\hat{s}$ and $\hat{\ell} = -\hat{d}^\perp = -\frac{\sin\tau^*}{|\sin\tau^*|}\hat{n}$, alternating with the stroke direction. The mean aerodynamic force components are

$$\begin{aligned}\langle F_n^* \rangle &= -\frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \langle C_L \sin\tau^* |\sin\tau^*| \rangle \\ &= \frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 C_{L,0} \langle \sin 2\psi \ \ \sin\tau^* |\sin\tau^*| \rangle \end{aligned} \tag{33}$$

$$\begin{aligned}\langle F_s^* \rangle &= \frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \langle C_D \sin\tau^* |\sin\tau^*| \rangle \\ &= \frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \langle \cos 2\psi \ \ \sin\tau^* |\sin\tau^*| \rangle \end{aligned} \tag{34}$$

Note the constant drag term vanishes because $\langle \sin\tau^* |\sin\tau^*| \rangle = 0$. We introduce the shorthand $\Delta C_D = C_{D,\frac{\pi}{2}} - C_{D,0}$. The expressions feature a common pre-factor we call q^*, defined as

$$q^* = \frac{1}{4}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \tag{35}$$

We further introduce the normal and in-plane force coefficients $C_{F_n^*}$ and $C_{F_s^*}$, defined as

$$C_{F_n^*} = \frac{F_n^*}{q^*} = 2C_{L,0}\langle \sin 2\psi \ \ \sin\tau^* |\sin\tau^*| \rangle \tag{36}$$

$$C_{F_s^*} = \frac{F_s^*}{q^*} = \Delta C_D \langle \cos 2\psi \ \ \sin\tau^* |\sin\tau^*| \rangle \tag{37}$$

which we synthesize into the overall force magnitude coefficient

$$C_{F^*} = \sqrt{C_{F_n^*}^2 + C_{F_s^*}^2} \tag{38}$$

This coefficient is useful because it is only a function of the pitch kinematic parameters and the lift and drag coefficient parameters, and is independent of morphology and wingbeat frequency and amplitude, which appear only in the pre-factor q^*. By maximizing the value of this coefficient, given a constraint on the force direction required to achieve a given maneuver, the characteristic wingbeat velocity $s_0^*\omega^*$ can be minimized. With our parametrization of the pitch kinematics, and with the lift and drag coefficient parameters set, C_{F^*} and its (n, s) decomposition are functions of 3 variables: ψ_0, ψ_1, and δ_0. We visualize C_{F^*}, $C_{F_n^*}$, and $C_{F_s^*}$ across slices of this 3D space in Figure 4 and Figure 5.

As panel (a) of Figure 5 makes evident, significantly more force is produced when $\delta_0 = \pm 90°$. Because $\delta_0 = -90°$ results in the mean force pointing along $-\hat{n}$ when the mean pitch is neutral ($\psi_0 = 0°$), we select $\delta_0 = +90°$ as the operating point. This choice maximizes the mean force for a given (ψ_0, ψ_1), and reduces the pitch parameter space to 2D. How then do we select ψ_0 and ψ_1? It is clear from panel (b) that a certain value of ψ_1, around $\psi_1 \approx 50°$, maximizes C_{F^*} for any given ψ_0. We will call it ψ_1^{opt}. In addition, $\psi_0 = 0°$ (mod $90°$) maximizes C_{F^*}. Looking at panel (b) of Figure 4, it is clear that the in-plane force component is zero when $\psi_0 = 0°$ (mod $90°$), so the force is purely normal to the stroke plane. In addition, from panel (a) of the same figure, we see that $\psi_0 = 0°$ results in the mean force pointing towards $+\hat{n}$, while $\psi_0 = \pm 90°$ results in the mean force pointing towards $-\hat{n}$. We select neutral pitch ($\psi_0 = 0°$) as the reference, for consistency with the choice of $\delta_0 = 90°$. In summary, the reference pitch kinematics

$$\psi_0 = 0° \qquad \psi_1 = \psi_1^{\text{opt}} \qquad \delta_0 = 90° \tag{39}$$

result in $C_{F^*} = C_{F^*}^{\max}$, and $\langle \vec{F}^* \rangle$ pointing towards $+\hat{n}$. These reference kinematics allow for *efficient* hover (in the sense that the characteristic wingbeat velocity $s_0^*\omega^*$ is minimized, because C_{F^*} is maximized) when the

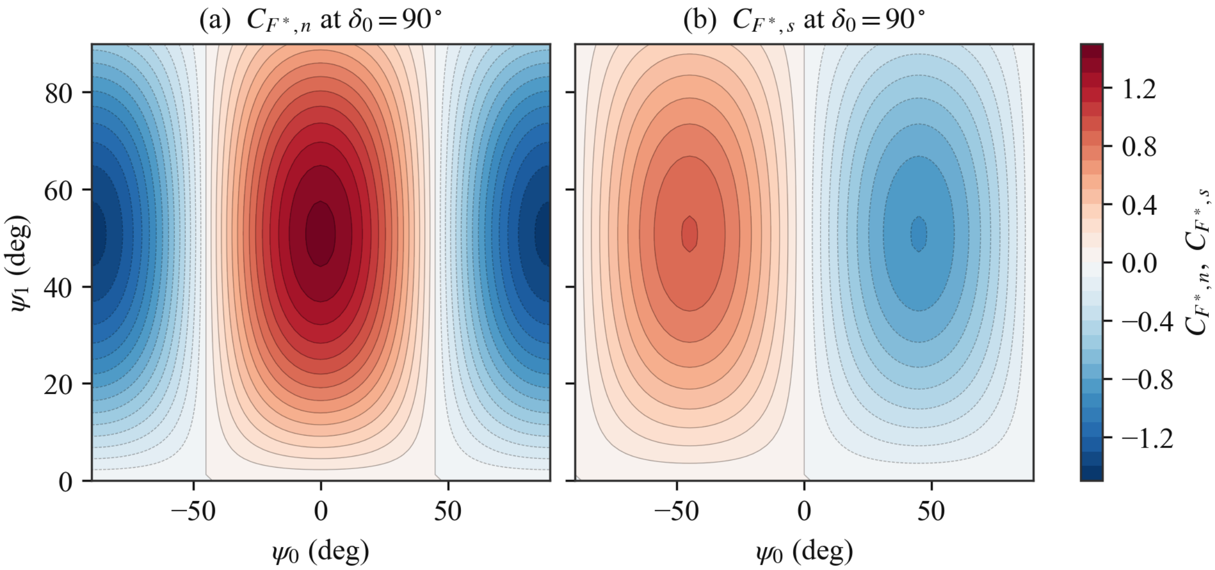

Figure 4: (a) Stroke-plane-normal (lift) component $C_{F_n^*}$ and (b) in-plane (drag) component $C_{F_s^*}$ of the force coefficient over the pitch plane at $\delta_0 = 90°$. The force direction quadrant in the $(\hat{s}, \hat{n})$ plane can be inferred from the signs.

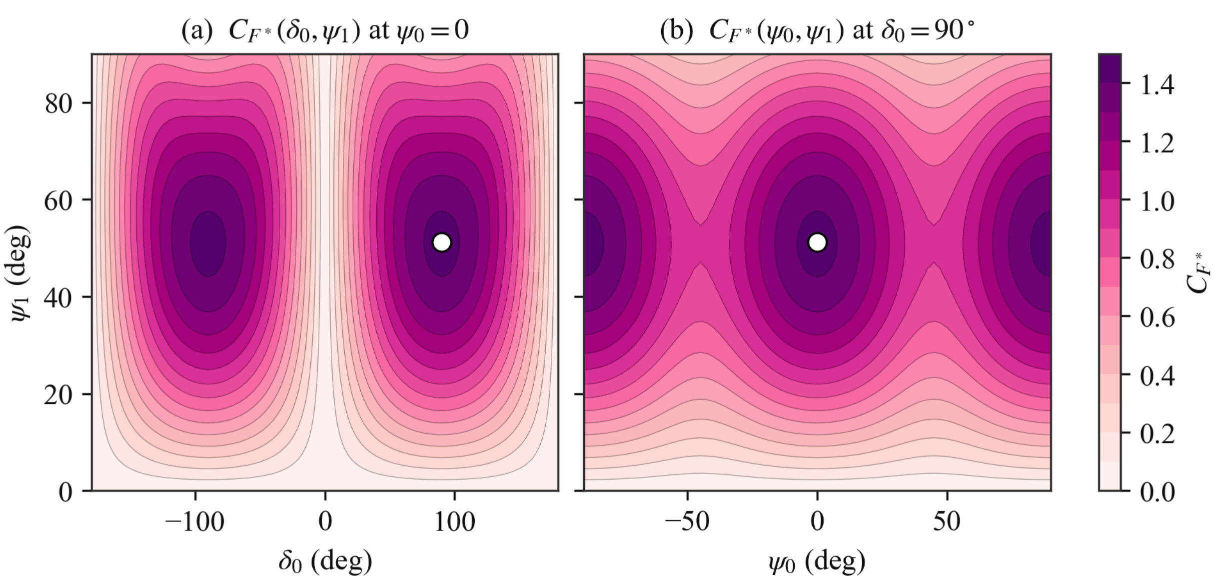

Figure 5: Dependence of C_{F^*} on pitch parameters at zero body velocity. Maximum shown as white dots.

stroke plane is horizontal ($\gamma^{\text{hover}} = 0°$). Indeed, since the mean force vector points towards $+\hat{n}$, $+\hat{n}$ must be vertical and opposed to the direction of gravity. To model dragonfly hover specifically, at an inclined stroke plane ($\gamma^{\text{hover}} \neq 0°$), the mean force vector must have an in-plane component. Looking at panel (b) of Figure 4, it is clear that this can be achieved by changing ψ_0, away from the symmetric 0° point. For convenience, we introduce β, the signed angle from the stroke-plane normal $\hat{n}$ to the mean force vector. Since the mean force is $\langle \vec{F}^* \rangle = q^* \left(C_{F_n^*} \hat{n} + C_{F_s^*} \hat{s} \right)$, β can be expressed from the $C_{F_n^*}$ and $C_{F_s^*}$ coefficients (Equation 36 and Equation 37) as

$$\beta = -\tan^{-1} \frac{C_{F_s^*}}{C_{F_n^*}} = -\tan^{-1} \left[\frac{\Delta C_D}{2C_{L,0}} \frac{\langle \cos 2\psi \sin \tau^* |\sin \tau^*| \rangle}{\langle \sin 2\psi \sin \tau^* |\sin \tau^*| \rangle} \right] \tag{40}$$

We can simplify the $\langle \cos 2\psi \sin \tau^* |\sin \tau^*| \rangle / \langle \sin 2\psi \sin \tau^* |\sin \tau^*| \rangle$ ratio in the above expression. With $\delta_0 = 90°$, the pitch kinematics reduce to $\psi = \psi_0 + \psi_1 \sin \tau^*$. Using the appropriate trigonometric identities,

$$\cos 2\psi = \cos 2\psi_0 \cos (2\psi_1 \sin \tau^*) - \sin 2\psi_0 \sin (2\psi_1 \sin \tau^*) \tag{41}$$

$$\sin 2\psi = \sin 2\psi_0 \cos (2\psi_1 \sin \tau^*) + \cos 2\psi_0 \sin (2\psi_1 \sin \tau^*) \tag{42}$$

Under the half-period shift $\tau^* \to \tau^* + \pi$, the factors $\sin \tau^* |\sin \tau^*|$ and $\sin (2\psi_1 \sin \tau^*)$ change sign while $\cos (2\psi_1 \sin \tau^*)$ is unchanged, so the $\cos (2\psi_1 \sin \tau^*)$ terms average to zero, leading to

$$\langle \cos 2\psi \;\; \sin \tau^* |\sin \tau^*| \rangle = -\sin 2\psi_0 \langle \sin (2\psi_1 \sin \tau^*) \sin \tau^* |\sin \tau^*| \rangle \tag{43}$$

$$\langle \sin 2\psi \;\; \sin \tau^* |\sin \tau^*| \rangle = \cos 2\psi_0 \langle \sin (2\psi_1 \sin \tau^*) \sin \tau^* |\sin \tau^*| \rangle \tag{44}$$

The common $\langle \sin (2\psi_1 \sin \tau^*) \sin \tau^* |\sin \tau^*| \rangle$ factor cancels out in the ratio, giving simply

$$\frac{\langle \cos 2\psi \;\; \sin \tau^* |\sin \tau^*| \rangle}{\langle \sin 2\psi \;\; \sin \tau^* |\sin \tau^*| \rangle} = -\tan 2\psi_0 \tag{45}$$

and Equation 40 becomes

$$\beta = \tan^{-1} \left[\frac{\Delta C_D}{2C_{L,0}} \tan 2\psi_0 \right] \tag{46}$$

Importantly, β is independent of ψ_1, and is only controlled by ψ_0. It follows that ψ_0 is an appropriate control variable for the mean force direction relative to the stroke-plane normal $\hat{n}$. β is shown as a function of ψ_0 in Figure 6. As discussed above, C_{F^*} becomes less than its maximum when $\psi_0 \neq 0° \pmod{90°}$, away from the reference pitch kinematics, so it is also shown in addition to β. In the $\Delta C_D / (2C_{L,0}) \to 1$ limit, the relation between β and ψ_0 can be approximated as $\beta \approx 2\psi_0$. This holds only roughly here because $\Delta C_D / (2C_{L,0}) = 1.9/3 \approx 0.63$, but the trend is the same, as can be seen in the figure. To achieve hover at an inclined stroke plane with tilt angle γ^{hover}, the mean force vector angle must be $\beta = -\gamma^{\text{hover}}$. This requires $\psi_0 = \psi_0^{\text{hover}}$, given by Equation 46 (specifically the inverse relation) when $\beta = -\gamma^{\text{hover}}$.

Now that we have narrowed down the pitch kinematics for hover, we step back to consider the flapping kinematics as well, governed by parameters s_0^* (amplitude) and ω^* (frequency). The condition for hover on the mean force magnitude is

$$C_{F^*} q^* = 1 \implies (s_0^* \omega^*)^2 = \frac{4}{C_{F^*} \frac{A^*}{m^*}} \tag{47}$$

We prefer to consider ω^* as a fixed parameter rather than a control variable, leaving s_0^* as the primary control for the force amplitude. The hover value is given by

$$(s_0^*)^{\text{hover}} = \frac{1}{\omega^*} \sqrt{\frac{4}{C_{F^*}^{\text{hover}} \frac{A^*}{m^*}}} \tag{48}$$

The hover kinematic parameters are summarized in Table 1. The control variables are s_0^* and ψ_0.

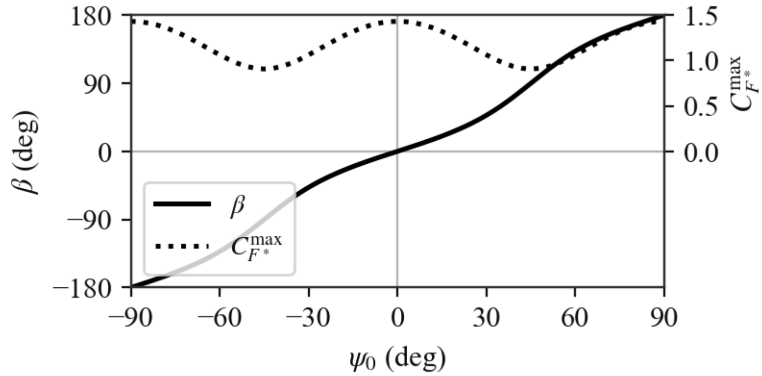

Figure 6: Effect of ψ_0 on force direction angle β and force coefficient C_{F^*}.

Table 1: Wing kinematic parameters for hover

Parameter	Value					Variables
s_0^*	$(s_0^*)^{\text{hover}}$	$\frac{A^*}{m^*}$	ω^*	γ^{hover}	$C_{L,0}$	ΔC_D
ψ_0	ψ_0^{hover}			γ^{hover}	$C_{L,0}$	ΔC_D
ψ_1	ψ_1^{opt}					None
δ_0	90°					None

3.2.3 General Case ($J \geq 0$)

For $J \neq 0$, the wing velocity relative to the air is no longer purely in-plane, so lift and drag each contribute to both force components. In terms of the angle χ, the wing velocity direction is $\hat{v} = \cos\chi\,\hat{n} - \sin\chi\,\hat{s}$. Drag acts along $-\hat{v}$, and lift acts along the perpendicular direction

$$\hat{\ell} = -\hat{d}^{\perp} = -\cos\chi\,\hat{s} - \sin\chi\,\hat{n} \tag{49}$$

recovering the hover directions at $\chi = \pm\frac{\pi}{2}$. The instantaneous force is

$$\vec{F}^* = \frac{1}{2}\frac{A^*}{m^*}(v^*)^2\left(C_L\,\hat{\ell} - C_D\,\hat{v}\right) \tag{50}$$

with components

$$F_n^* = -\frac{1}{2}\frac{A^*}{m^*}(v^*)^2\,(\sin\chi\,C_L + \cos\chi\,C_D) \tag{51}$$

$$F_s^* = -\frac{1}{2}\frac{A^*}{m^*}(v^*)^2\,(\cos\chi\,C_L - \sin\chi\,C_D) \tag{52}$$

Substituting $\cos\chi = J/\sqrt{J^2 + \sin^2\tau^*}$, $\sin\chi = \sin\tau^*/\sqrt{J^2 + \sin^2\tau^*}$, and the expression for $(v^*)^2$ (Equation 27), we get

$$F_n^* = -\frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2\sqrt{J^2 + \sin^2\tau^*}\;(\sin\tau^*\,C_L + J\,C_D) \tag{53}$$

$$F_s^* = -\frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2\sqrt{J^2 + \sin^2\tau^*}\;(J\,C_L - \sin\tau^*\,C_D) \tag{54}$$

From $\alpha = \psi - \chi$ (Equation 29), the aerodynamic coefficients (Equation 1 and Equation 2) expand as

$$C_L = C_{L,0} \left(\cos 2\chi \ \sin 2\psi - \sin 2\chi \ \cos 2\psi\right) \tag{55}$$

$$C_D = \bar{C}_D - \frac{\Delta C_D}{2} \left(\cos 2\chi \ \cos 2\psi + \sin 2\chi \ \sin 2\psi\right) \tag{56}$$

where we define $\bar{C}_D \equiv \left(C_{D,\frac{\pi}{2}} + C_{D,0}\right)/2$ for brevity, and where the double angles are smooth functions of the wing phase:

$$\cos 2\chi = \frac{J^2 - \sin^2 \tau^*}{J^2 + \sin^2 \tau^*} \qquad \sin 2\chi = \frac{2J \sin \tau^*}{J^2 + \sin^2 \tau^*} \tag{57}$$

Substituting into the force components and averaging, the force coefficients are

$$\begin{aligned} C_{F_n^*} = & -2 \left\langle \frac{\left[C_{L,0}\left(J^2 - \sin^2 \tau^*\right) - \Delta C_D \, J^2\right] \sin \tau^* \ \sin 2\psi}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle \\ & + 2 \left\langle \frac{J\left[2C_{L,0} \sin^2 \tau^* + \frac{\Delta C_D}{2}\left(J^2 - \sin^2 \tau^*\right)\right] \cos 2\psi}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle - 2J\bar{C}_D \left\langle \sqrt{J^2 + \sin^2 \tau^*} \right\rangle \end{aligned} \tag{58}$$

$$\begin{aligned} C_{F_s^*} = & -2 \left\langle \frac{J\left[C_{L,0}\left(J^2 - \sin^2 \tau^*\right) + \Delta C_D \sin^2 \tau^*\right] \sin 2\psi}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle \\ & + 2 \left\langle \frac{\left[2C_{L,0} J^2 - \frac{\Delta C_D}{2}\left(J^2 - \sin^2 \tau^*\right)\right] \sin \tau^* \ \cos 2\psi}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle \end{aligned} \tag{59}$$

Note the constant ($\bar{C}_D$) drag term of $C_{F_s^*}$ vanishes because $\langle \sin \tau^* \sqrt{J^2 + \sin^2 \tau^*} \rangle = 0$. Setting $J = 0$ recovers Equation 36 and Equation 37, as expected. At the operating point $\delta_0 = 90°$, $\psi = \psi_0 + \psi_1 \sin \tau^*$, and the parity argument of the hover case applies unchanged: under $\tau^* \to \tau^* + \pi$, the bracketed polynomials and $\sqrt{J^2 + \sin^2 \tau^*}$ are invariant, so expanding $\sin 2\psi$ and $\cos 2\psi$ and discarding the vanishing averages leaves

$$C_{F_n^*} = 2\, I_n(J, \psi_1) \ \cos 2\psi_0 - 2J\bar{C}_D \, I_0(J) \tag{60}$$

$$C_{F_s^*} = -2\, I_s(J, \psi_1) \ \sin 2\psi_0 \tag{61}$$

with

$$I_0 = \left\langle \sqrt{J^2 + \sin^2 \tau^*} \right\rangle \tag{62}$$

$$\begin{aligned} I_n = & \left\langle \frac{J\left[2C_{L,0} \sin^2 \tau^* + \frac{\Delta C_D}{2}\left(J^2 - \sin^2 \tau^*\right)\right] \cos\left(2\psi_1 \sin \tau^*\right)}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle \\ & - \left\langle \frac{\left[C_{L,0}\left(J^2 - \sin^2 \tau^*\right) - \Delta C_D \, J^2\right] \sin \tau^* \ \sin\left(2\psi_1 \sin \tau^*\right)}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle \end{aligned} \tag{63}$$

$$\begin{aligned} I_s = & \left\langle \frac{J\left[C_{L,0}\left(J^2 - \sin^2 \tau^*\right) + \Delta C_D \sin^2 \tau^*\right] \cos\left(2\psi_1 \sin \tau^*\right)}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle \\ & + \left\langle \frac{\left[2C_{L,0} J^2 - \frac{\Delta C_D}{2}\left(J^2 - \sin^2 \tau^*\right)\right] \sin \tau^* \ \sin\left(2\psi_1 \sin \tau^*\right)}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle \end{aligned} \tag{64}$$

The angle from the stroke-plane normal to the mean force (Equation 40) becomes

$$\beta = \tan^{-1}\left[\frac{I_s \sin 2\psi_0}{I_n \cos 2\psi_0 - J\bar{C}_D\, I_0}\right] \tag{65}$$

In the limit $J \to 0$, the I_0 term vanishes and

$$I_n \to C_{L,0}\,\langle \sin\left(2\psi_1 \sin\tau^*\right) \sin\tau^* |\sin\tau^*|\rangle \qquad I_s \to \frac{\Delta C_D}{2}\,\langle \sin\left(2\psi_1 \sin\tau^*\right) \sin\tau^* |\sin\tau^*|\rangle \tag{66}$$

recovering Equation 46. The overall force coefficient C_{F^*} is shown in Figure 7 for values of J from 0 to 1, at $\delta_0 = 90°$. Highlighted on each panel is the reference point from the hover plot Figure 5, tracked as it turns from a global maximum at $J = 0$ to a local maximum, and eventually to a saddle point. This point represents the maximum force available in the $+\hat{n}$ direction for a given J. The region of space within the dotted contour around the reference point corresponds to $C_{F_n^*} > 0$. Within this region, the stroke-plane-normal component of the mean force points towards $+\hat{n}$.

The values of C_{F^*} and ψ_1 at the reference point are shown as functions of J in Figure 8. As the figure makes clear, to obtain the maximum force along $+\hat{n}$, ψ_1 should be decreased as J increases. The mechanism behind this is illustrated in Figure 9. For a given ψ, α decreases as J is increased. This moves us away from the ideal α, reducing lift, and in turn the force component along $+\hat{n}$. By decreasing ψ, the ideal α is recovered, thrust along $+\hat{n}$ is maximized. Of course, this is an instantaneous picture, but the conclusion can be extended to the mean force over the wingbeat stroke. To achieve the required reduction in ψ everywhere throughout the stroke, it is hopefully clear that the relevant parameter is ψ_1.

Finally, Figure 10 shows β as a function of ψ_0 for J from 0 to 1. Specifically, since β now depends on ψ_1 in addition to ψ_0 (see Equation 65), each line is obtained using the ψ_1 value adapted to the value of J (as shown in Figure 7). As J is increased, there is a clear steepening of the slope around $\psi_0 = 0$, meaning control authority is significantly increased. This would tend to suggest that any gain associated with the control variable ψ_0 may need to adapt based on the current J.

We summarize the kinematic parameters for the general case in Table 2. Since there is no requirement for steady flight like in hover, no specific values for the control variables s_0^* and ψ_0 are prescribed, however the relevant variables that would inform the choice of the values are still listed.

Table 2: Wing kinematic parameters (general case)

Parameter	Value	Variables
s_0^*	—	$\frac{A^*}{m^*}$ ω^* γ $C_{L,0}$ ΔC_D $\bar{C}_D$ J
ψ_0	—	γ $C_{L,0}$ ΔC_D $\bar{C}_D$ J
ψ_1	$\psi_1^{\text{opt}}(J)$	J $C_{L,0}$ ΔC_D
δ_0	90°	None

3.2.4 Constant-Pitch Simplification

The general-case results above are exact but not closed-form: ψ_1 remains inside the cycle averages through the $\cos\left(2\psi_1 \sin\tau^*\right)$ and $\sin\left(2\psi_1 \sin\tau^*\right)$ factors, so quantities such as $\psi_1^{\text{opt}}(J)$ and the maps of Figure 7 must be evaluated numerically. Closed forms can be obtained by replacing the sinusoidal pitch schedule with the constant-pitch idealization of section 2, restated here in terms of the stroke-direction sign $\sigma \equiv -\sin\tau^*/|\sin\tau^*|$ ($\sigma = +1$ on the $+\hat{s}$ half-stroke):

$$\psi(\tau^*) = \psi_0 - \sigma\,\psi_1 \tag{67}$$

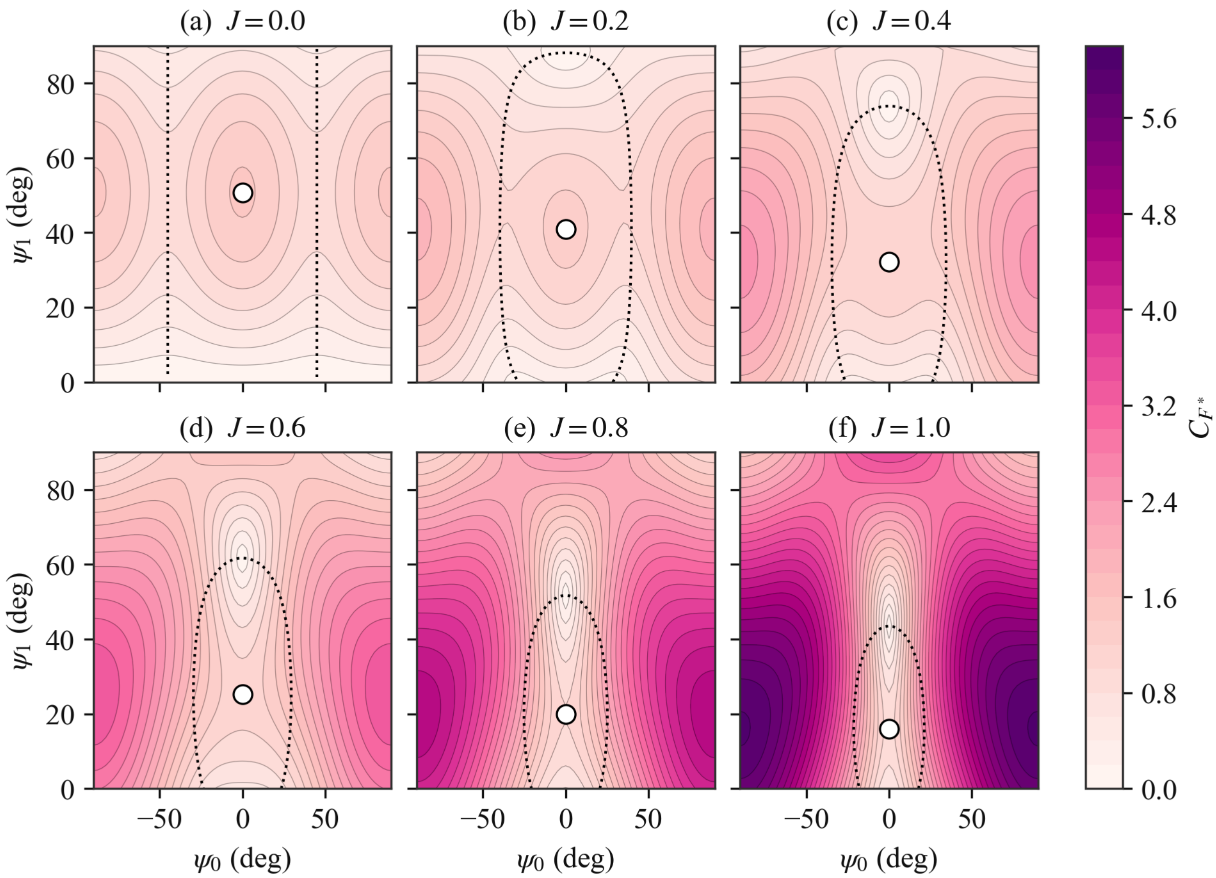

Figure 7: Force coefficient C_{F^*} over the pitch plane (ψ_0, ψ_1) at $\delta_0 = 90°$, for increasing advance ratio J. White dots track the hover optimum as it moves with J. It remains on the $\psi_0 = 0°$ axis by symmetry while ψ_1 decreases, and it turns from a local maximum into a saddle point. The global maximum sits at large $|\psi_0|$, where the force is dominated by drag from the normal inflow and points along $-\hat{n}$. The dotted contour marks $C_{F_n^*} = 0$, the boundary between propulsive $(+\hat{n})$ and braking $(-\hat{n})$ mean force.

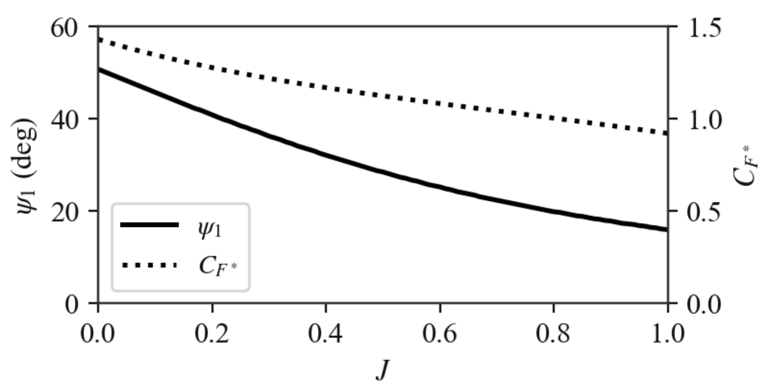

Figure 8: Pitch amplitude ψ_1 and force coefficient C_{F^*} of the continued hover optimum (the point tracked in Figure 7) as functions of advance ratio J. The point turns from a local maximum into a saddle at $J \approx 0.46$.

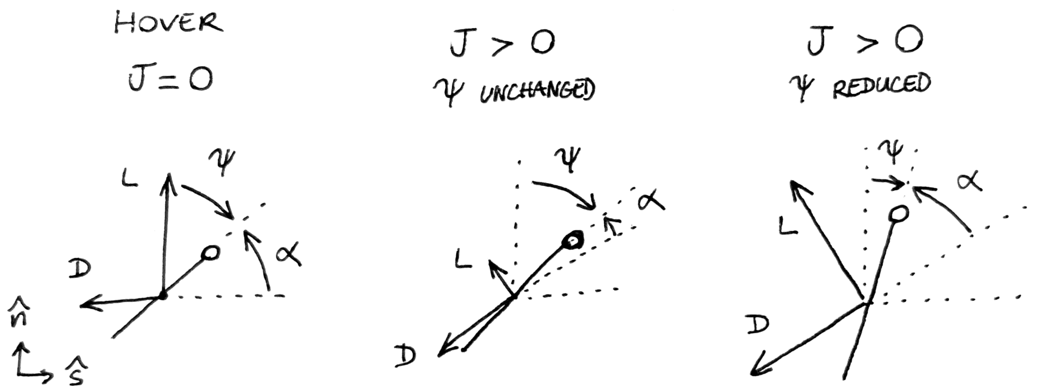

Figure 9: Illustration of the ψ reduction mechanism for maximizing stroke-plane-normal force as J increases.

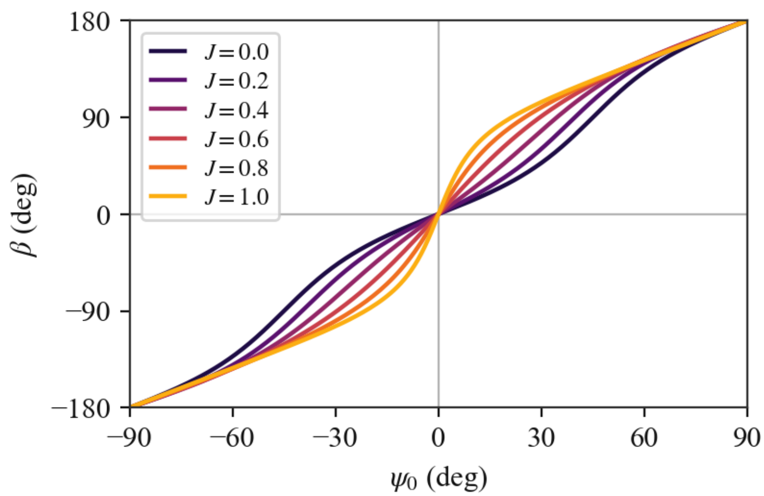

Figure 10: Effect of ψ_0 on the force direction angle β for increasing advance ratio J, at $\delta_0 = 90°$ and with ψ_1 set, for each J, to the continued-optimum value of Figure 8. Unlike in the hover case, β depends on ψ_1 when $J > 0$.

which holds the pitch constant at $\psi_0 - \psi_1$ during the forward half-stroke ($\sigma = +1$) and at $\psi_0 + \psi_1$ during the backward half-stroke ($\sigma = -1$). The pitch flip occurs at stroke reversal, since we impose $\delta_0 = 90°$.

This implies instantaneous rotation of the wing, which is not achievable by a real wing but is mostly harmless in the quasi-steady model because the in-plane wing velocity vanishes at reversal. The flip produces at most a bounded force discontinuity due to the normal inflow, when $J > 0$. The simulations in the remainder of this report retain the sinusoidal schedule; the constant-pitch law serves purely as an analytical surrogate.

Since $\sigma^2 = 1$, the pitch trigonometric factors are piecewise constant over each half-stroke:

$$\sin 2\psi = \sin 2\psi_0 \cos 2\psi_1 - \sigma \cos 2\psi_0 \sin 2\psi_1 \tag{68}$$

$$\cos 2\psi = \cos 2\psi_0 \cos 2\psi_1 + \sigma \sin 2\psi_0 \sin 2\psi_1 \tag{69}$$

Comparing with the expansions used in the hover case, the constant-pitch law simply amounts to the substitutions

$$\cos(2\psi_1 \sin\tau^*) \to \cos 2\psi_1 \qquad \sin(2\psi_1 \sin\tau^*) \to -\sigma \sin 2\psi_1 \tag{70}$$

Since σ changes sign under the half-period shift $\tau^* \to \tau^* + \pi$, exactly like the $\sin(2\psi_1 \sin\tau^*)$ factor it replaces, every vanishing average identified previously still vanishes, and the pitch dependence now factors entirely out of the remaining averages.

In the hover case, applying the substitutions to Equation 36 and Equation 37, and using $\langle \sigma \sin\tau^* |\sin\tau^*|\rangle = -\langle \sin^2\tau^*\rangle = -\frac{1}{2}$, gives

$$C_{F_n^*} = C_{L,0} \sin 2\psi_1 \cos 2\psi_0 \qquad C_{F_s^*} = -\frac{\Delta C_D}{2} \sin 2\psi_1 \sin 2\psi_0 \tag{71}$$

and therefore

$$C_{F^*} = \sin 2\psi_1 \sqrt{C_{L,0}^2 \cos^2 2\psi_0 + \frac{\Delta C_D^2}{4} \sin^2 2\psi_0} \tag{72}$$

The structure observed numerically in Figure 4 and Figure 5 is now explicit. The component ratio $C_{F_s^*}/C_{F_n^*} = -\left(\Delta C_D/2C_{L,0}\right)\tan 2\psi_0$ reproduces Equation 46 identically—as it must, since that relation was derived independently of the ψ_1 schedule. The pitch amplitude enters only through the $\sin 2\psi_1$ prefactor, so the optimum is exact and independent of ψ_0:

$$\psi_1^{\text{opt}} = 45^\circ \qquad C_{F^*}^{\max} = C_{L,0} \tag{73}$$

the latter attained at the reference $\psi_0 = 0^\circ$ (since $C_{L,0} > \Delta C_D/2$). The maximum has a transparent interpretation: with $\psi_1 = 45^\circ$ and $\psi_0 = 0^\circ$, the angle of attack is $\alpha = \psi + \sigma\frac{\pi}{2} = \sigma\, 45^\circ$, i.e. $|\alpha| = 45^\circ$ on both half-strokes—the constant-pitch law holds the wing at the lift-maximizing angle of attack for the entire cycle, so the force coefficient equals $C_{L,0}$ exactly. The sinusoidal schedule only passes through this ideal angle twice per half-stroke, which is why its optimum is displaced ($\psi_1^{\text{opt}} \approx 51^\circ$) and slightly weaker ($C_{F^*}^{\max} \approx 1.43$ versus $C_{L,0} = 1.5$, a 5% overestimate by the constant-pitch law).

In the general case, the same substitutions pull ψ_1 out of I_n and I_s:

$$I_n = A_n \cos 2\psi_1 + B_n \sin 2\psi_1 \qquad I_s = A_s \cos 2\psi_1 + B_s \sin 2\psi_1 \tag{74}$$

where A_n, B_n, A_s, and B_s are functions of J only. Introducing the family of averages

$$W_k(J) = \left\langle \frac{|\sin\tau^*|^k}{\sqrt{J^2 + \sin^2\tau^*}} \right\rangle \qquad k \in \{0, 1, 2, 3\} \tag{75}$$

they read

$$A_n = J\left[\left(2C_{L,0} - \frac{\Delta C_D}{2}\right) W_2 + \frac{\Delta C_D}{2} J^2 W_0\right] \qquad A_s = J\left[C_{L,0} J^2 W_0 - \left(C_{L,0} - \Delta C_D\right) W_2\right] \tag{76}$$

$$B_n = C_{L,0} W_3 - \left(C_{L,0} - \Delta C_D\right) J^2 W_1 \qquad B_s = \frac{\Delta C_D}{2} W_3 + \left(2C_{L,0} - \frac{\Delta C_D}{2}\right) J^2 W_1 \tag{77}$$

The odd-k members are elementary,

$$W_1 = \frac{2}{\pi} \tan^{-1}\frac{1}{J} \qquad W_3 = \frac{1}{\pi}\left[\left(1 - J^2\right)\tan^{-1}\frac{1}{J} + J\right] \tag{78}$$

while the even-k members, together with I_0, are complete elliptic integrals: with $\mu = 1/\left(1 + J^2\right)$,

$$W_0 = \frac{2}{\pi}\sqrt{\mu}\, K(\mu) \qquad W_2 = I_0 - J^2 W_0 \qquad I_0 = \frac{2}{\pi}\frac{E(\mu)}{\sqrt{\mu}} \tag{79}$$

where $K(\mu) = \int_0^{\pi/2} \left(1 - \mu \sin^2 \theta\right)^{-1/2} d\theta$ and $E(\mu) = \int_0^{\pi/2} \left(1 - \mu \sin^2 \theta\right)^{1/2} d\theta$. Equation 60, Equation 61, and Equation 65 then hold unchanged with these I_n and I_s, so the mean aerodynamic force is closed-form in all of the kinematic parameters. Setting $J = 0$ gives $A_n = A_s = 0$, $B_n = C_{L,0}/2$, and $B_s = \Delta C_D/4$, recovering Equation 71.

Along the reference axis $\psi_0 = 0°$, Equation 60 becomes a pure harmonic in $2\psi_1$,

$$C_{F_n^*} = 2\left(A_n \cos 2\psi_1 + B_n \sin 2\psi_1\right) - 2J\bar{C}_D\, I_0 \tag{80}$$

so the continued hover optimum tracked numerically in Figure 8 becomes exact:

$$\psi_1^{\text{opt}}(J) = \frac{1}{2}\tan^{-1}\frac{B_n}{A_n} \qquad \left(C_{F_n^*}\right)^{\max} = 2\sqrt{A_n^2 + B_n^2} - 2J\bar{C}_D\, I_0 \tag{81}$$

At $J = 0$, $A_n = 0$ and $\psi_1^{\text{opt}} = 45°$; as J grows, A_n increases from zero (both A_n and B_n are positive here), so ψ_1^{opt} decreases monotonically. The closed form thus reproduces, and explains, the trend of Figure 8: the growing $A_n \cos 2\psi_1$ term, which carries the normal-inflow contributions and peaks at $\psi_1 = 0°$, pulls the optimum toward smaller pitch amplitudes—exactly the mechanism illustrated in Figure 9. Table 3 compares the constant-pitch predictions with the sinusoidal schedule: ψ_1^{opt} is offset low by 5–6°, and the available force is captured to within 5% across $J \in [0, 1]$ (an overestimate at hover, a slight underestimate at speed).

Table 3: Continued hover optimum at $\psi_0 = 0°$: constant-pitch closed form (Equation 81) versus the sinusoidal schedule evaluated numerically (Figure 8).

	ψ_1^{opt}		$\left(C_{F_n^*}\right)^{\max}$	
J	Constant	Sinusoidal	Constant	Sinusoidal
0	45.0°	50.8°	1.500	1.431
0.25	32.9°	38.5°	1.241	1.247
0.5	23.0°	28.5°	1.084	1.124
0.75	16.1°	21.1°	0.975	1.024
1	11.6°	16.0°	0.875	0.921

Beyond its transparency, this limit has practical value for control. The trim of the two control variables $\left(s_0^*, \psi_0\right)$ to a demanded mean force—performed numerically in subsection 3.4—becomes analytically invertible: for a demanded force direction, Equation 65 reduces to a harmonic equation of the form $a \sin 2\psi_0 + b \cos 2\psi_0 = c$ with known coefficients, yielding ψ_0 explicitly, after which s_0^* follows from the demanded force magnitude through q^* (Equation 35). The constant-pitch expressions could therefore serve as an analytical initialization—or replacement—for the numerical trim solver, at the cost of the few-percent model error quantified in Table 3.

3.3 Control Strategy

From the above analysis, we propose a control strategy. It will be discussed qualitatively in this section, and formalized in the following section (subsection 3.4). We begin by going over *what to do* with each of the wing kinematic parameters.

- *Stroke plane angle γ:* when hovering, $\gamma = \gamma^{\text{hover}}$. When moving, γ should change to effectively obtain, or at least approximate, the axial inflow we have assumed in the analysis. There is the issue of the transition between these two phases, which may be abrupt if the commanded flight direction is not aligned with the stroke-plane normal at hover. We propose that γ be varied smoothly from its hover value to its moving flight value based on J.
- *Stroke amplitude s_0^*:* primary control for the mean force magnitude, should be varied so that the force magnitude demand is met. In hovering or in general steady flight, this means having the mean force magnitude equal the weight.
- *Mean pitch ψ_0:* primary control for the mean force direction relative to the stroke-plane normal, should be varied so that the force direction demand is met. In hovering or in general steady flight, this means having the mean force vector point toward $+\hat{z}$.
- *Pitch amplitude ψ_1:* should be varied with J to maximize the available force, as shown in Figure 8.
- *Pitch phase δ_0:* fixed to 90°, not a control variable.
- *Wingbeat frequency ω^*:* constant parameter, not a control variable.

In summary, the two variables that directly act to modulate and direct the force are s_0^* and ψ_0. The two remaining variables, γ and ψ_1, adapt to the velocity vector to respectively approximate an axial inflow, and maximize the available force.

A comment should be made here about the $\hat{x}$-wise asymmetry introduced by the nonzero γ^{hover}. At low speeds, when the stroke plane is still close to its hover orientation, this will result in different behaviors for motion toward $+\hat{x}$ and $-\hat{x}$. To restore symmetry, we propose to flip the sign of γ^{hover} when the commanded flight direction changes.

3.4 Control Law

We now formalize the strategy of subsection 3.3. Let $\vec{x}^*$ and $\vec{u}^*$ be the body position and velocity. In general, a prescribed trajectory is characterized at every point by a reference position $\vec{x}_r^*$, velocity $\vec{u}_r^*$, and acceleration $\vec{a}_r^*$ (*e.g.* centripetal acceleration for a circular trajectory). Depending on the exact requirement for the controller, only some of these inputs may be necessary for the control law. In the case of hovering for example, if the objective is to hold a specific position in space, $\vec{x}_r^*$ will be needed, but $\vec{u}_r^*$ is only needed to provide additional damping through derivative control. If the objective is simply to hover about an arbitrary point, then only $\vec{u}_r^*$ is needed. Similarly, in the case of prey pursuit, the input is the line-of-sight bearing to the prey, without range information, and the objective is to align the velocity vector with the line of sight as we close in on the prey. In this case, only $\vec{u}_r^*$ needs to be formulated, which prescribes both the pursuit direction (line of sight) and the desired closing speed. With this in mind, we start by forming the force demand in the most general case, using all the potential inputs. The result is a proportional-derivative tracking law with acceleration feedforward and compensation of the weight

$$\vec{a}_{\text{des}}^* = \vec{a}_r^* + K_p \left(\vec{x}_r^* - \vec{x}^*\right) + K_d \left(\vec{u}_r^* - \vec{u}^*\right) \tag{82}$$

$$\langle \vec{F}^* \rangle^{\text{des}} = \vec{a}_{\text{des}}^* + \hat{z} \tag{83}$$

In the hover and prey pursuit cases, $\vec{a}_r^* = \vec{0}$. In the prey pursuit case specifically, $K_p = 0$, while either K_p or K_d may be zero in the hover case. We will see that in practice, a simpler velocity-based expression ($\vec{a}^*_{\text{des}} = K_d \left(\vec{u}_r^* - \vec{u}^*\right)$) is very satisfactory. With the force demand formulated, γ and ψ_1 must now adapt to the measured velocity. The advance ratio is evaluated as

$$J = \frac{|\vec{u}^*|}{\left(s_0^* \omega^*\right)^{\text{ref}}} \tag{84}$$

where $\left(s_0^*\omega^*\right)^{\text{ref}}$ is the wingbeat velocity of the reference configuration at the hover condition, held fixed so that J does not depend on the control variable s_0^*. The stroke-plane angle blends from the inclined-hover value to the velocity-aligned value γ_u, defined such that $\hat{n}$ is aligned with $\vec{u}^*$:

$$\gamma = [1 - f(J)]\, \sigma_x \gamma^{\text{hover}} + f(J)\, \gamma_u \qquad\qquad f(J) = 1 - e^{-(J/J_c)^2} \tag{85}$$

with $J_c = 0.35$, and where $\sigma_x \in \{-1, +1\}$ is the sign of the commanded horizontal heading, latched at its most recent nonzero value so that the stroke plane does not flip on the residual hover velocity (see subsection 3.3). The pitch amplitude is slaved to the force-maximizing value of Figure 8,

$$\psi_1 = \psi_1^{\text{opt}}(J) \tag{86}$$

under an assumed axial inflow. The actual inflow deviates from the stroke-plane normal during transients, but only appreciably at low advance ratio, where the optimum is insensitive to the inflow direction, so the deviation is neglected. With (γ, ψ_1) now set, the remaining two variables (s_0^*, ψ_0) are trimmed to the force demand by solving

$$\langle \vec{F}^* \rangle \left(s_0^*, \psi_0;\, \gamma, \psi_1, J\right) = \langle \vec{F}^* \rangle^{\text{des}} \tag{87}$$

for (s_0^*, ψ_0), using the expressions for $\langle \vec{F}^* \rangle$ derived in subsection 3.2. Finally, once the reference trajectory is exhausted and the position error falls within a small radius (0.3 body lengths), γ and ψ_1 are pinned to their hover values $\sigma_x \gamma^{\text{hover}}$ and $\psi_1^{\text{opt}}(0)$, avoiding jitter from the residual hover velocity, while Equation 87 continues to hold hover equilibrium.

4 Results

4.1 Reference Configuration

To ground our results, we introduce a reference configuration based on morphological and wingbeat frequency data for *Anisoptera* from [4] and [5], respectively. Nondimensionalized, the data range is [0.2, 1.0] for A^*/m^*, and [8, 20] for ω^*. For the reference configuration, we select the intermediate values

$$\frac{A^*}{m^*} = 0.6 \qquad\qquad \omega^* = 14 \tag{88}$$

In addition, we choose the hovering stroke plane angle to be

$$\gamma^{\text{hover}} = -45° \tag{89}$$

defined with a negative sign (clockwise) so that it corresponds to a horizontal dragonfly pointing towards $+\hat{x}$. When settling from lateral motion towards $-\hat{x}$, the hovering stroke plane angle would be $-\gamma^{\text{hover}} = +45°$, as explained in subsection 3.3.

Since dragonflies have two wing pairs, we set $N = 2$, with the two pairs flapping in antiphase (their phase parameters ϕ differing by 180°) so as to reduce body oscillations.

4.2 Hover

The hover regime of the control law is exercised by initializing the body at the reference position, $\vec{x}^*(0) = \vec{x}_r^* = (0, 0)$, with a velocity perturbation $\vec{u}^*(0) = (0.2, 0.2)$ and $\vec{u}_r^* = 0$. Position feedback only ($K_d = 0$), velocity feedback only ($K_p = 0$), and the full tracking law are compared in Figure 11, Figure 12, and Figure 13, with $K_p = 2.25$ and $K_d = 2.7$.

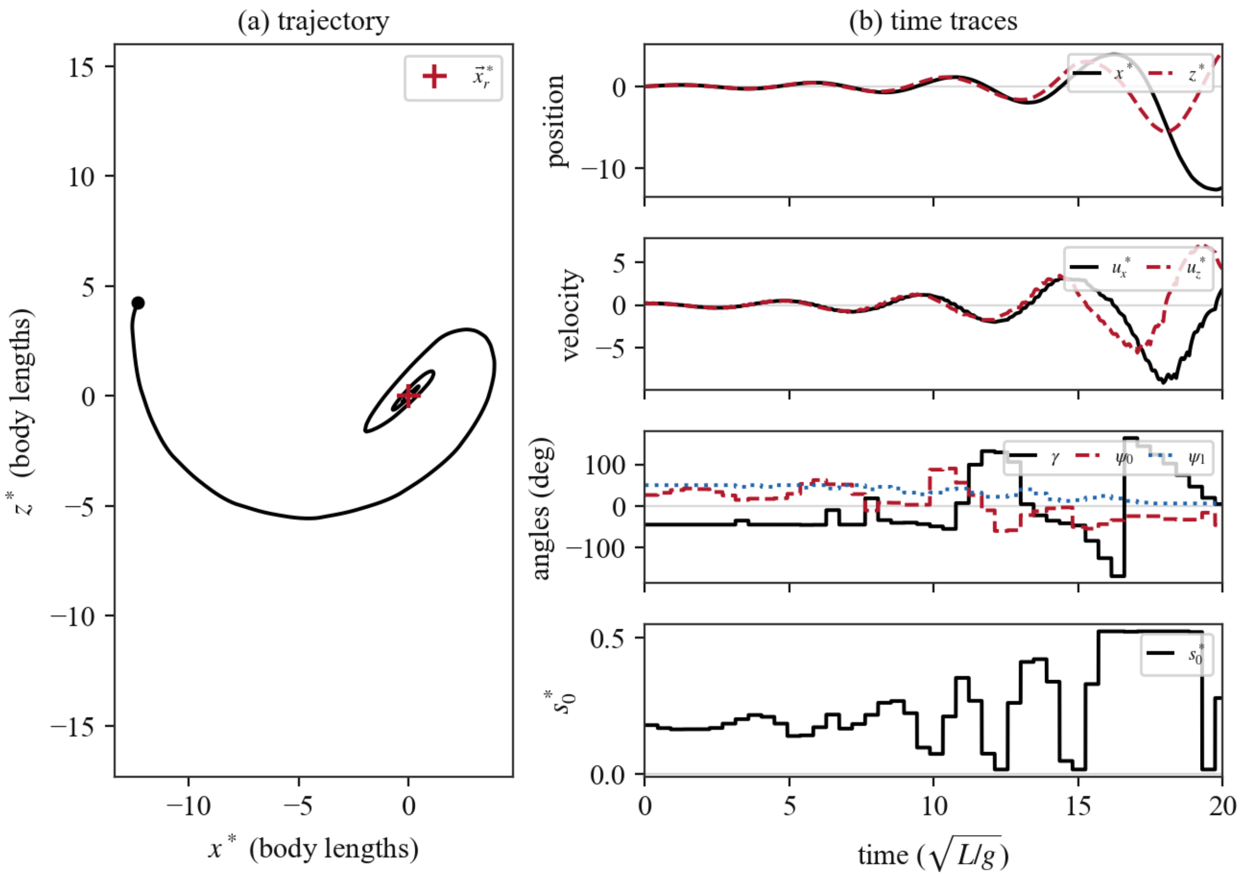

Figure 11: Hover hold with position feedback only ($K_d = 0$). The oscillation about $\vec{x}_r^*$ is not damped, and the phase lag of the once-per-wingbeat control update makes it grow. Once the position error exceeds the hover pin radius, γ and ψ_1 resume scheduling and the trim saturates.

4.3 Prescribed Trajectory

We choose an upward bend as the reference trajectory: a horizontal run of 1.5 body lengths, a circular arc of radius 2 bending up by 45°, and a straight climb of 1.5 body lengths, traversed with a trapezoidal speed schedule (linear ramps from and to rest over a distance of 0.3, cruise speed 0.7). The reference acceleration $\vec{a}_r^*$ carries the tangential (ramp) and centripetal (arc) demand. Starting from rest on the path, four configurations are compared: position feedback only (Figure 14), velocity feedback only (Figure 15), both (Figure 16), and both with acceleration feedforward—the full law of Equation 83 (Figure 17)—again with $K_p = 2.25$ and $K_d = 2.7$.

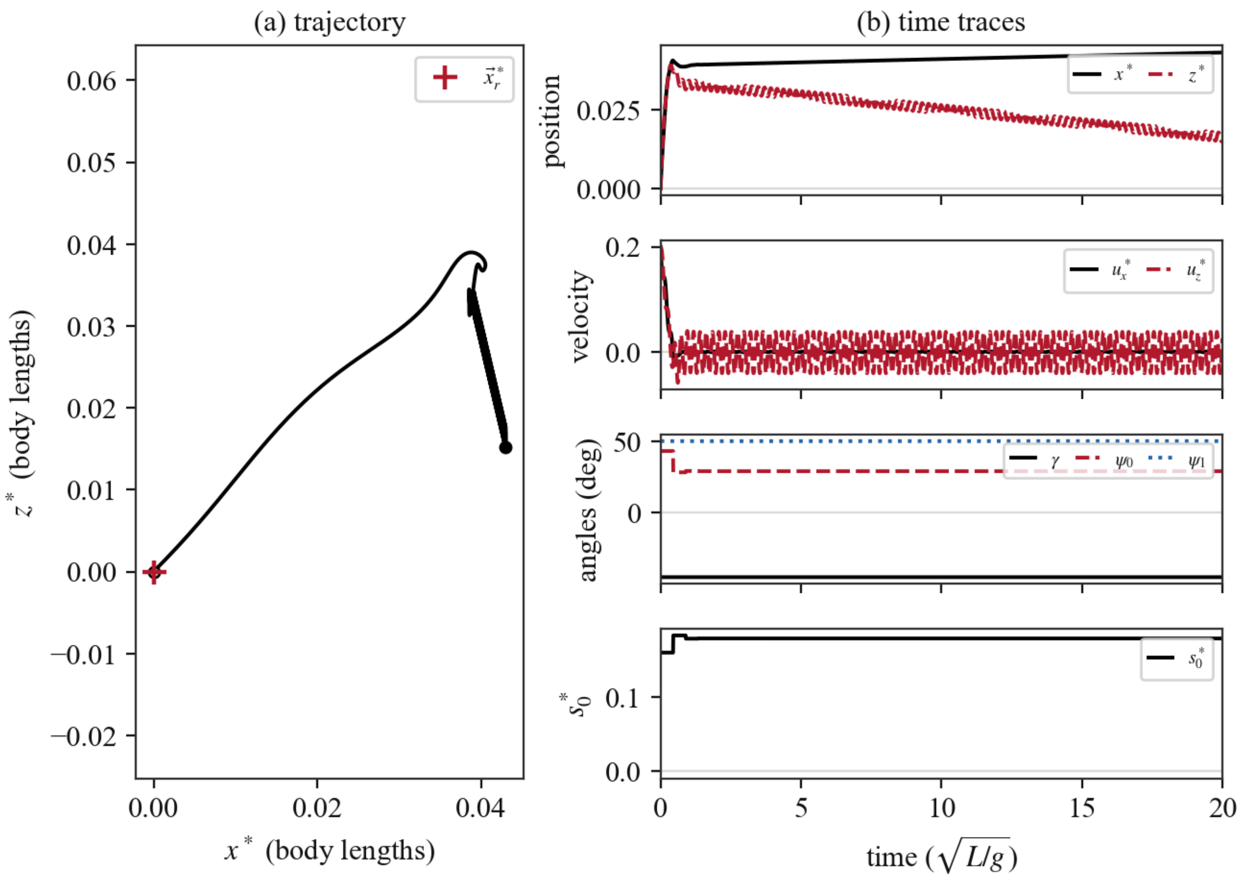

Figure 12: Hover hold with velocity feedback only ($K_p = 0$). The velocity perturbation is eliminated within about one time unit, but the position offset accumulated meanwhile (about 0.05 body lengths) is never corrected. With that said, the resulting error is very small, and while there is some residual drift over time, the result is quite satisfactory.

4.4 Prey Pursuit

Prey pursuit closes the guidance loop through the control law using only the velocity term of Equation 83. The reference velocity is defined as the desired closing speed along the line of sight to the prey,

$$\vec{u}_r^* = u_{\text{cmd}}^* \, \hat{r} \qquad \hat{r} = \frac{\vec{x}_{\text{prey}}^* - \vec{x}^*}{|\vec{x}_{\text{prey}}^* - \vec{x}^*|} \tag{90}$$

with no position reference or acceleration feedforward, so that $\langle \vec{F}^* \rangle^{\text{des}} = K_d(\vec{u}_r^* - \vec{u}^*) + \hat{z}$. The prey starts at $(2.5, 2.0)$ and moves at the constant velocity $(0.35, 0)$; the body starts at rest at the origin, with $u_{\text{cmd}}^* = 0.7$ and $K_d = 2.7$. Interception is declared at a range of 0.5 body lengths. The resulting pursuit is shown in Figure 18.

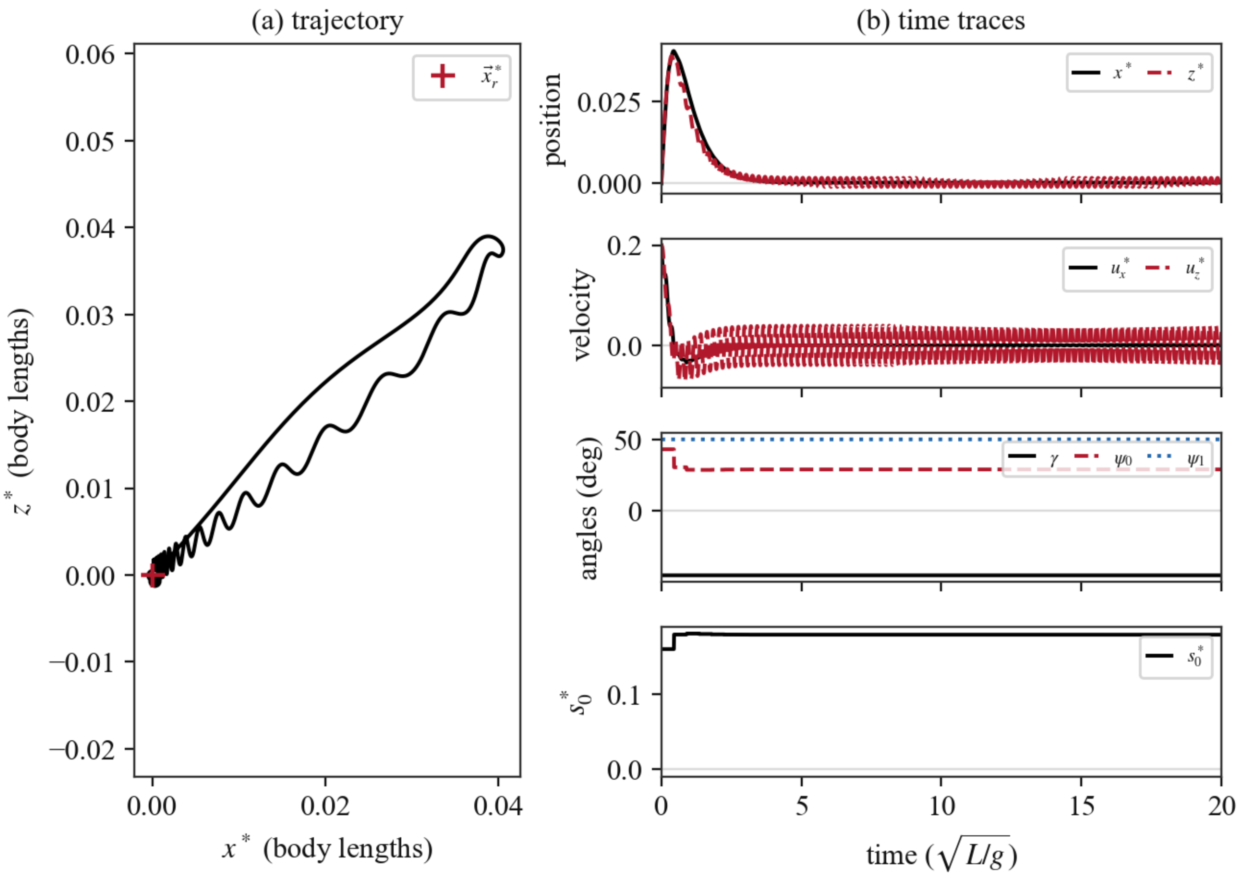

Figure 13: Hover hold with the full tracking law. The body returns to $\vec{x}_r^*$ (peak excursion 0.06 body lengths, residual position error below 2×10^{-3}).

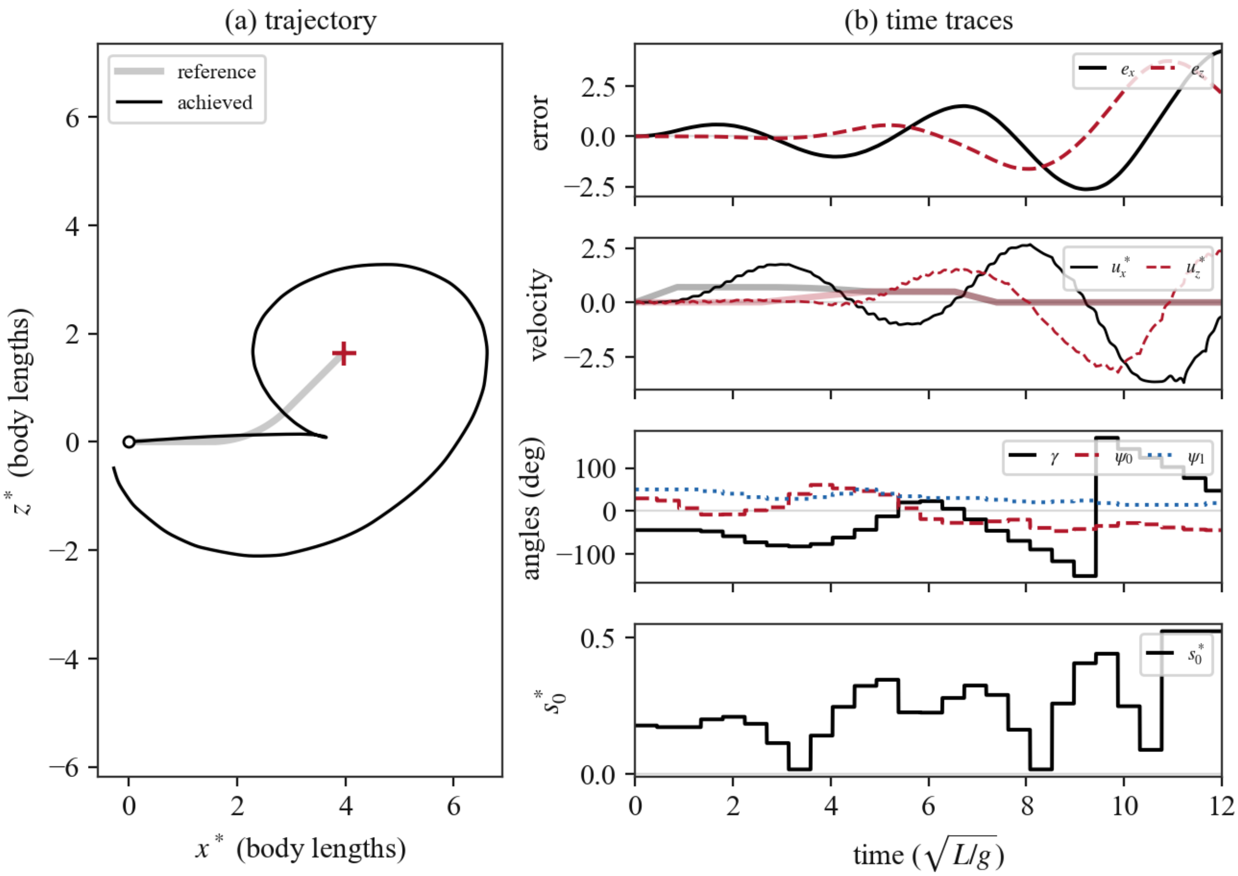

Figure 14: Prescribed trajectory with position feedback only ($K_d = 0$): as in hover, the undamped error dynamics under the once-per-wingbeat update are unstable, and the body departs from the path.

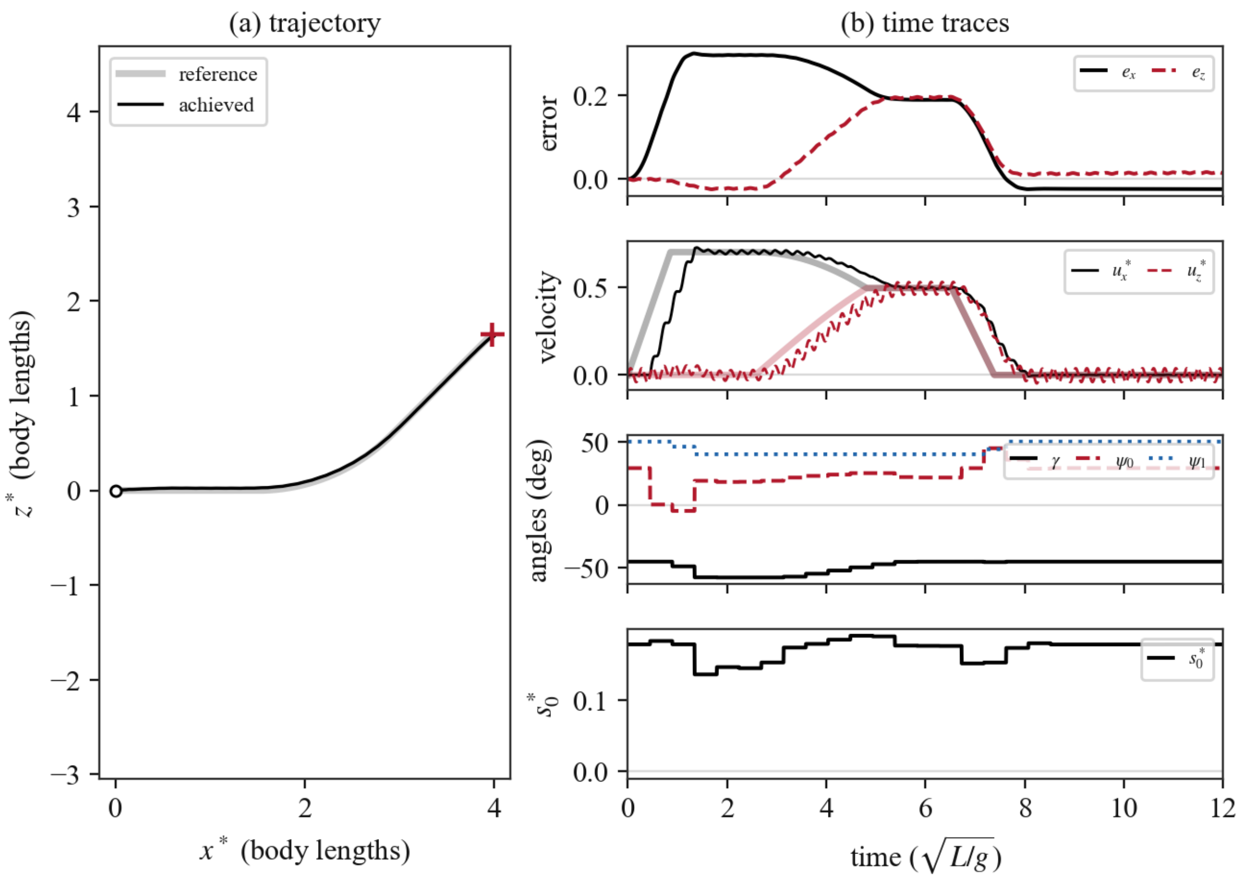

Figure 15: Prescribed trajectory with velocity feedback only ($K_p = 0$): the body follows the commanded velocity with a first-order lag, trailing the reference by $|\vec{u}_r^*|/K_d \approx 0.26$ body lengths throughout. The deceleration ramp lets it catch up near the end, but nothing corrects position drift. With that said, the resulting trajectory appears very satisfactory.

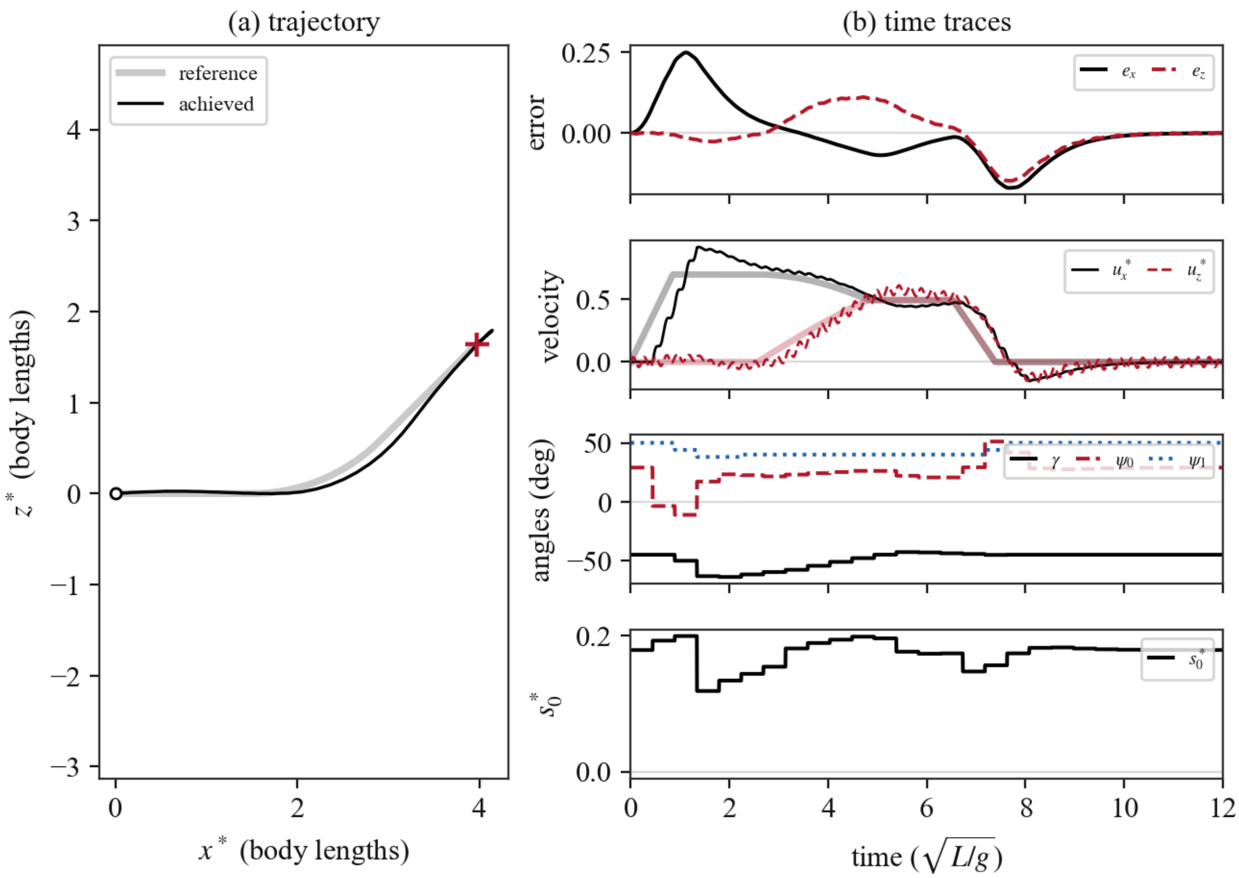

Figure 16: Prescribed trajectory with position and velocity feedback ($K_p + K_d$, no feedforward). Tracking error peaks at 0.25 body lengths, dominated by the speed-ramp lag and by the "cutting" inside the arc, since the required acceleration must be generated by a standing error.

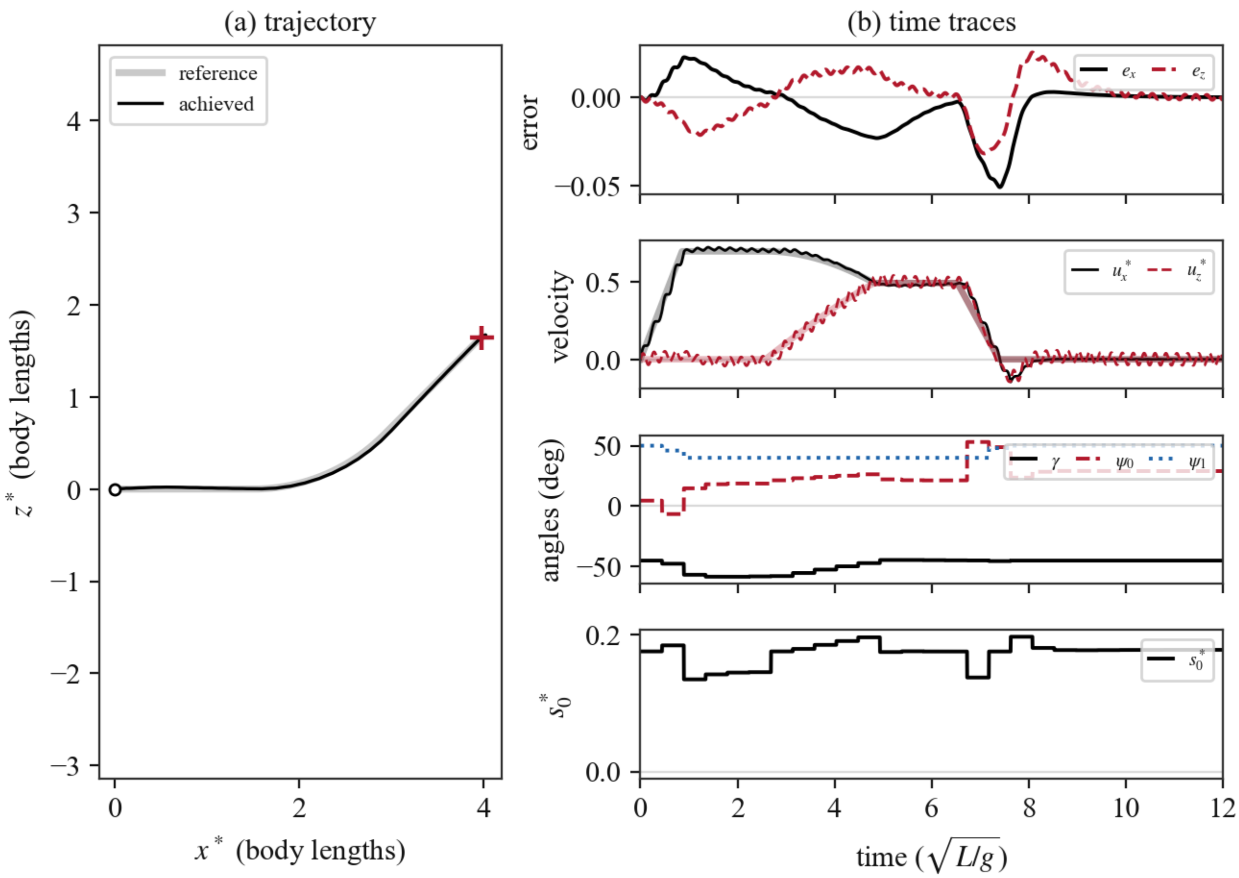

Figure 17: Prescribed trajectory with the full law of Equation 83, including the acceleration feedforward $\vec{a}_r^*$. The maneuver demand is supplied directly rather than through the error terms, and the peak tracking error drops significantly.

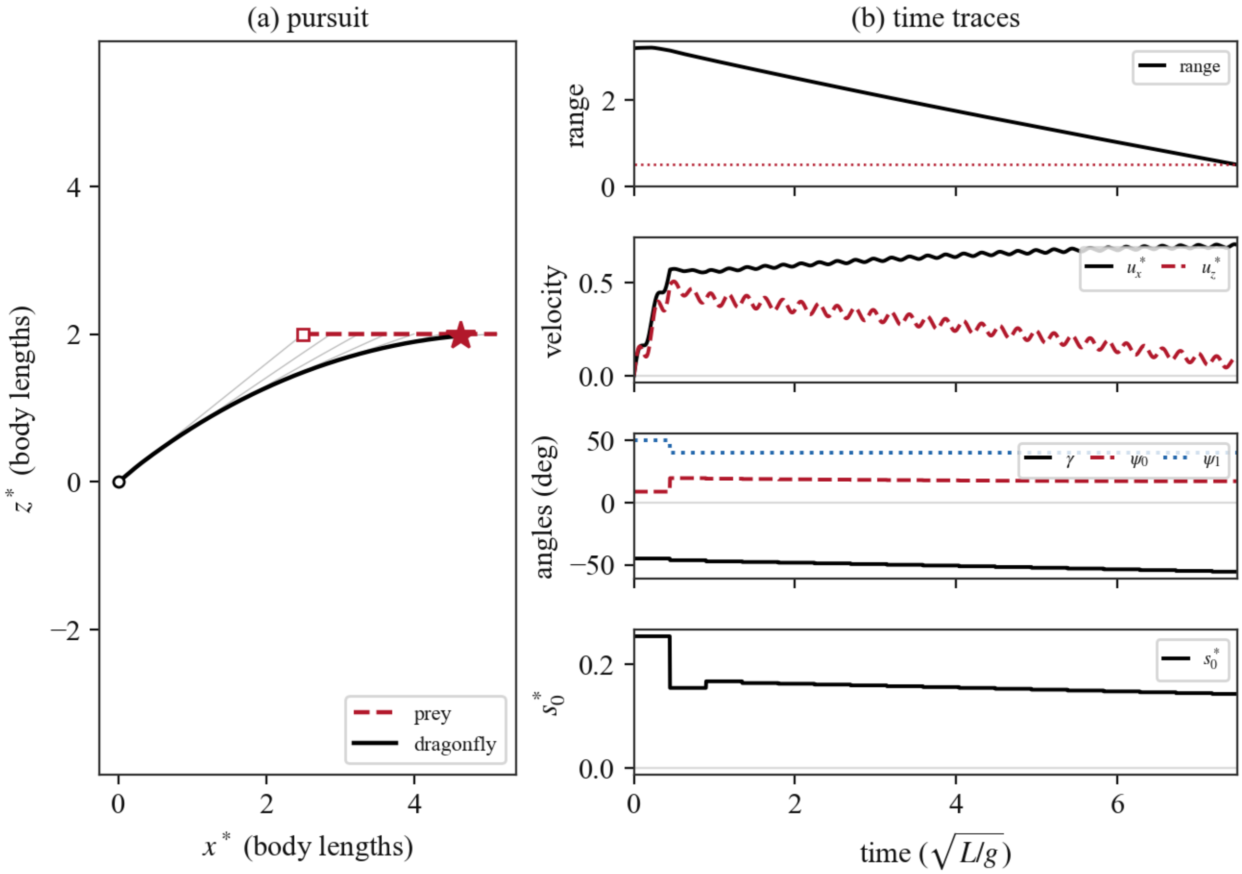

Figure 18: Prey pursuit. (a) The line-of-sight command produces the classical pure-pursuit tail chase, with capture at $t^* \approx 7.5$ (star). (b) The body accelerates to the commanded closing speed ($J \approx 0.2$). γ leans from γ^{hover} toward alignment with the flight direction, ψ_1 follows its slaved schedule (Equation 86), and (s_0^*, ψ_0) trim the force demand.

5 Conclusion

The proposed control scheme is capable of achieving hover and prey pursuit, and of following arbitrary trajectories to a reasonable degree of accuracy. Some caveats are worth mentioning:

- *Not strictly proportional control*. While all control tasks can be achieved with a single proportional gain K_d in the force demand formulation, the control variables vary nonlinearly with the force demand, since we are solving the inverse problem of which variable values satisfy the force demand. Following the hover controller presented in the previous report, we may be able to resolve this by linearizing the force, using the expressions we derived in subsection 3.2.

- *Issue with descent*. When descending (towards $-\hat{z}$) at a fast rate, the stroke-plane normal will point downwards. However, to achieve a steady descent velocity, the force vector must point up, opposed to gravity. The force vector, then, must point toward $-\hat{n}$. This can be achieved as we saw in subsection 3.2 with large enough $|\psi_0|$, but at high J it results in large force magnitudes which lead to instability in the unsteady dynamics, and prevent clean descent trajectories. This could potentially be addressed with an alternate control mode using gliding instead of flapping to achieve steady descent. Note that descent is not as problematic if the commanded trajectory has a sustained downward acceleration larger than that due to gravity, since the requirement that the force oppose gravity does not hold.

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